ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Incommensurable Variable

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did toy algebra tiles from 1990 make x = 3, and why did that ruin student understanding?'
- Ask: 'Can you ever combine 2x + 3 into 5x? Why do the physical pieces prevent that?'
- Ask: 'What does the area of a tile represent algebraically?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to force 3.5 units to equal x. Explain that x is a continuous variable that can represent any real number.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If x was 10, how many unit tiles would balance one x-rod in weight? If x was 100?"
""

OFFICIAL SOLUTION KEY:

If x=10, 10 units. If x=100, 100 units. The physical rod represents the variable independent of its assignment.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Incommensurable Variable

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why can't you just count unit squares to find the value of x? In this lab, you prove that x is physically incapable of being an integer.

STEP-BY-STEP APPARATUS SETUP:

1. Pull one x-rod (51.0 × 14.0 mm) and several unit tiles (14.0 × 14.0 mm) from your kit.
2. Try to line up unit tiles along the length of the x-rod.
3. Notice: Three unit tiles (42.0 mm) are too short; four unit tiles (56.0 mm) are too long!

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Algebra Tiles: 1 x^2 flat (51×51mm), 1 x-rod (51×14mm), unit tiles (14×14mm)
DATUM TARGET:	x = 51.0 mm, Unit = 14.0 mm -> Ratio = 3.6428... != Integer (Kills x=2 Fallacy)
CALCULATOR KEY:	[x] [ENTER] [3] [+] -> x + 3

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7A, Chapter 4 (Algebraic Expressions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Variables & Patterns)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 3 • CCSS.MATH.7.EE.A.1 • TEKS 7.10A
GRADE LEVEL BAND:	Grades 7-9 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 3 ('Algebra & Transcendentals'), pp. 48-62
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 2 (Incommensurability Proof) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE INCOMMENSURABILITY PROOF

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Measure: 3 units = 42.0 mm. 4 units = 56.0 mm. The x-rod is exactly 51.0 mm.

Step 2: Conclude: x is NOT equal to 3, and x is NOT equal to 4. It cannot be counted in units!

Step 3: Now compare the width of the x-rod to a unit tile: both are exactly 14.0 mm.

Step 4: This means an x-rod has dimensions 1 unit wide by x units long, giving it an AREA of exactly 1 × x = x.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
x = 51.0 mm, Unit = 14.0 mm –> Ratio = 3.6428... != Integer (Kills x=2 Fallacy)

PHYSICAL APPARATUS MANIFEST:

Algebra Tiles: 1 x^2 flat (51×51mm), 1 x-rod (51×14mm), unit tiles (14×14mm)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE x^2 FLAT VERIFICATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Place the large square tile (x^2 flat) on your desk. Prove that its area is exactly x × x.

FORM A MASTER STATION TASKS (Complete All 3):

1 **Place an x-rod along the horizontal edge of the x^2 flat.**
Expected Physical Finding: **Length matches exactly (51.0 mm)**

2 **Place a second x-rod along the vertical edge of the x^2 flat.**
Expected Physical Finding: **Height matches exactly (51.0 mm)**

3 **What is the algebraic area of this tile?**
Expected Physical Finding: **Area = x · x = x^2**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x] [ENTER] [3] [+] –> x + 3

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x	X: x	Variable operand x
ENTER	Y: x | X: x	Stack lift
3	Y: x | X: 3	Constant operand 3
+	X: x + 3	Expression evaluated symbolically

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: The Distributive Area Model

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does multiplying x by x produce a square tile?'
- Ask: 'How does this area model prove that x(a + b) = xa + xb?'
- Ask: 'What would happen if you tried to fill the corner with unit tiles instead of an x^2 flat?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding lengths to areas (e.g., claiming the area is x + x + 3). Remind them: boundary rails show LENG TH; interior tiles show AREA.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A rectangle has an area of $3x^2 + 6x$. If its width is 3x, what is its physical length?"
""

OFFICIAL SOLUTION KEY:

Length = $(3x^2 + 6x) / 3x = x + 2$.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: The Distributive Area Model

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Multiplying algebra terms is not just shuffling letters on a page—it is building the surface area of a physical rectangle.

STEP-BY-STEP APPARATUS SETUP:

1. Snap the two L-frame coordinate rails together at a 90° right angle on your desk.
2. Along the vertical axis, lay down one x-rod (length x).
3. Along the horizontal axis, lay down one x-rod and three unit tiles (length x + 3).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: L-Frame Coordinate Rails, 1 x^2 flat, 5 x-rods, 6 unit tiles

DATUM TARGET: Area Rectangle: Width = x, Height = (x + 3) -> Total Area = $x^2 + 3x$

CALCULATOR KEY: [x] [ENTER] [x] [3] [+] [*] -> x^2 + 3x

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7A, Chapter 5 (Algebraic Expansion)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6, Module 2 (Area Model for Multiplication)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 8, Ch. 7 • CCSS.MATH.HSA.APR.A.1 • TEKS A.10D
GRADE LEVEL BAND:	Grades 7–9 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 3, pp. 62–74
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 5 (L-Frame Coordinate Setup) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: FILLING THE COORDINATE STAGE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Your axes define a rectangle with dimensions x by (x + 3).

Step 2: Fill the interior corner: Place one x^2 flat at the corner (dimensions x by x).

Step 3: Fill the remaining space to the right: Place three x-rods side-by-side (each 1 by x).

Step 4: Count the physical pieces inside the rectangle: Exactly one x^2 tile and three x-rods! Total area = $x^2 + 3x$.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Area Rectangle: Width = x, Height = (x + 3) -> Total Area = $x^2 + 3x$

PHYSICAL APPARATUS MANIFEST:

L-Frame Coordinate Rails, 1 x^2 flat, 5 x-rods, 6 unit tiles

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: EXPANDING 2x(x + 2)

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Set the horizontal axis to (x + 2) and the vertical axis to 2x (two x-rods). Fill the rectangle.

FORM A MASTER STATION TASKS (Complete All 3):

1 **How many x^2 flats fit inside the frame?**
Expected Physical Finding: **Two x^2 flats ($2x^2$)**

2 **How many x-rods fit beside them?**
Expected Physical Finding: **Four x-rods (4x)**

3 **Write the expanded algebraic identity.**
Expected Physical Finding: **$2x(x + 2) = 2x^2 + 4x$**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: `[x] [ENTER] [x] [3] [+] [*] -> x^2 + 3x`

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x	X: x	Width operand
ENTER	Y: x | X: x	Stack lift
x+3	Y: x | X: x + 3	Height operand
*	X: $x^2 + 3x$	Multiply polynomials via distribution

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: Binomial Multiplication

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can the 3 horizontal rods and 2 vertical rods be added together into 5x?'
- Ask: 'Why can\'t the 6 unit tiles be added to the 5x rods?'
- Ask: 'How does this physical model make sign errors impossible?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking (x + 2)(x + 3) equals x² + 6 (forgetting the middle term). Point directly to the 5 x-rods oc
cupying the middle zones.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"Model (x + 4)(x + 1) using tiles. What are the 4 quadrant pieces, and what is the final polynomial?"
""

OFFICIAL SOLUTION KEY:

Quadrants: x², 1x, 4x, 4 units -> Simplified: x² + 5x + 4.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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step-by-step video solution, and printable rubric.

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: Binomial Multiplication

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does FOIL (First, Outside, Inside, Last) work? Because every binomial multiplication forms four physical quadrant zones in a rectangle.

STEP-BY-STEP APPARATUS SETUP:

1. Set the horizontal axis of the L-frame to (x + 3): one x-rod + three unit tiles.
2. Set the vertical axis of the L-frame to (x + 2): one x-rod + two unit tiles.
3. Prepare to fill the entire rectangular space enclosed by the rails.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: L-Frame Rails, 1 x² flat, 5 x-rods, 6 unit tiles

DATUM TARGET: Area: (x + 2)(x + 3) = x² + 3x + 2x + 6 = x² + 5x + 6

CALCULATOR KEY: [x] [2] [+] [ENTER] [x] [3] [+] [*] -> x^2 + 5x + 6

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 8A, Chapter 3 (Expansion of Binomials)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 5, Unit 6, Module 3 (Binomial Arrays)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Algebra 1, Ch. 7, Sec. 7.2 • CCSS.MATH.HSA.APR.A.1 • TEKS A.10B

GRADE LEVEL BAND: Grades 8–9 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 2: SC-32 Scientific Handbook, Ch. 7 ('Polynomials & Brent Solver'), pp. 88–96

HARDWARE UNBOXING BOOKLET: ALGB-01 Booklet: Page 6 (Aha! Discovery: FOIL Area Model)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE 4-ZONE CORNER GRID

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1 (First): Fill the top-left corner where x meets x: Place one x^2 flat.

Step 2 (Outside): Fill the top-right zone where x meets 3 units: Place three horizontal x-rods.

Step 3 (Inside): Fill the bottom-left zone where 2 units meet x: Place two vertical x-rods.

Step 4 (Last): Fill the bottom-right corner where 2 units meet 3 units: Place a 2×3 array of unit tiles (6 units).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Area: $(x + 2)(x + 3) = x^2 + 3x + 2x + 6 = x^2 + 5x + 6$

PHYSICAL APPARATUS MANIFEST:

L-Frame Rails, 1 x^2 flat, 5 x-rods, 6 unit tiles

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: COMBINING LIKE PIECES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Gather and group all the identical physical tiles inside the completed rectangle.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Count the x^2 flats.**
   Expected Physical Finding: 1 (x^2)
2. **Combine all the x-rods: 3 horizontal + 2 vertical.**
   Expected Physical Finding: 5 x-rods (5x)
3. **Count the unit tiles in the corner.**
   Expected Physical Finding: 6 unit tiles (+6)
4. **Write the final simplified polynomial.**
   Expected Physical Finding: $x^2 + 5x + 6$

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x] [2] [+] [ENTER] [x] [3] [+] [*] -> x^2 + 5x + 6

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x+2	X: x + 2	First binomial factor
ENTER	Y: x + 2 | X: x + 2	Stack lift
x+3	Y: x + 2 | X: x + 3	Second binomial factor
*	X: $x^2 + 5x + 6$	Polynomial multiplication

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 4: Factoring Quadratics

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did finding the factors depend on finding two numbers that add to 7 and multiply to 12?'
- Ask: 'What happens if you have a prime polynomial like $x^2 + 5x + 7$? Can it form a rectangle?'
- Ask: 'How does the physical puzzle eliminate guessing on tests?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to place unit tiles directly adjacent to the x^2 flat. Remind them: only edges of matching leng
th can touch!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Factor $x^2 + 8x + 15$ using tiles. How must the 8 rods be split to accommodate the 15 unit tiles?"
""

OFFICIAL SOLUTION KEY:

Split 8 into 3 and 5, since 3 × 5 = 15. Factors: $(x + 3)(x + 5)$.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 1: Factoring Quadratics

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Factoring is simply playing the area game in reverse: given a pile of loose tiles, can you assemble them into one perfect seamless rectangle?

STEP-BY-STEP APPARATUS SETUP:

1. Take 1 x^2 flat, 7 x-rods, and 12 unit tiles from your tray.
2. Your challenge: Arrange ALL 20 pieces into a solid, gap-free rectangle.
3. Rule: The x^2 flat must stay in the top-left corner; unit tiles must stay in the bottom-right.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Algebra Tiles: 1 x^2 flat, 7 x-rods, 12 unit tiles + L-Frame Rails

DATUM TARGET: Reconstruct Rectangle: Area $x^2 + 7x + 12$ -> Dimensions $(x + 3) \times (x + 4)$

CALCULATOR KEY: [x^2 + 7x + 12] [FACTOR] -> $(x + 3)(x + 4)$

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8A, Chapter 3, Lesson 3.3 (Factoring Quadratics)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Factoring Polynomial Arrays)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 1, Ch. 7, Sec. 7.5 • CCSS.MATH.HSA.SSE.A.2 • TEKS A.10E
GRADE LEVEL BAND:	Grades 8–10 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 7, pp. 96–104
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 6 (Factoring Reverse Puzzle)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 2: LAB INVESTIGATION: THE RECTANGLE RECONSTRUCTION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Place the x^2 flat at the corner.

Step 2: Try splitting the 7 x-rods: Place 6 along the top and 1 along the side. Can 12 unit tiles fill a 1×6 space? (No: 1×6=6, 6 units left over).

Step 3: Try splitting 7 into 5 and 2. Can 12 units fill a 2×5 space? (No: 2×5=10, 2 units left over).

Step 4: Try splitting 7 into 3 and 4! Notice: 3 × 4 = 12! The 12 unit tiles fit into a perfect 3×4 array with zero gaps!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Reconstruct Rectangle: Area $x^2 + 7x + 12$ -> Dimensions $(x + 3) \times (x + 4)$

PHYSICAL APPARATUS MANIFEST:

Algebra Tiles: 1 x^2 flat, 7 x-rods, 12 unit tiles + L-Frame Rails

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: READING THE DIMENSIONS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Lay the L-frame rails along the edges of your completed rectangle to read the factors.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Measure the horizontal edge of the rectangle.**
   Expected Physical Finding: **1 x-rod + 4 units = $(x + 4)$**
2. **Measure the vertical edge of the rectangle.**
   Expected Physical Finding: **1 x-rod + 3 units = $(x + 3)$**
3. **State the factored form of $x^2 + 7x + 12$.**
   Expected Physical Finding: **$(x + 3)(x + 4)$**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: `[x^2 + 7x + 12] [FACTOR] -> (x + 3)(x + 4)`

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
poly	X: $x^2 + 7x + 12$	Input quadratic polynomial
FACTOR	X: $(x + 3)(x + 4)$	Root decomposition into linear binomial factors

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: Zero-Pair Annihilation

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did the x-rods completely disappear from the final expression?'
- Ask: 'Why is this identity called the DIFFERENCE of squares?'
- Ask: 'Can you ever get a sum of squares like $x^2 + 4$ using real binomials?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking (−2) × (−2) equals −4. Point out that here we multiply (+2) by (−2), which produces negative tiles.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Factor $4x^2 - 9$. What are the binomial dimensions, and how many zero pairs are annihilated?"
""

OFFICIAL SOLUTION KEY:

Dimensions: (2x + 3)(2x − 3). Six (+x) rods and six (−x) rods annihilate (12 zero pairs).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: Zero-Pair Annihilation

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How can you multiply two binomials and end up with NO middle x-terms at all? In this lab, positive and negative tiles annihilate like matter and antimatter.

STEP-BY-STEP APPARATUS SETUP:

1. Pull positive tiles (mirror-ironed smooth) and negative tiles (micro-grooved ribbed).
2. Rule of Annihilation: When one positive (+1) and one negative (−1) touch, they form a 'Zero Pair' and vanish.
3. Set L-frame rails to (x + 2) vertically and (x − 2) horizontally.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: 1 x^2 flat, 2 positive x-rods (smooth), 2 negative x-rods (ribbed), 4 negative units

DATUM TARGET: Annihilation Invariant: (+x) + (−x) = 0 -> Net Area = $x^2 - 4$

CALCULATOR KEY: [x] [2] [+] [ENTER] [x] [2] [−] [*] -> x^2 − 4

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8A, Chapter 3, Lesson 3.4 (Difference of Squares)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Zero Pairs & Integer Operations)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 1, Ch. 7, Sec. 7.7 • CCSS.MATH.HSA.SSE.A.2 • TEKS A.10F
GRADE LEVEL BAND:	Grades 8–10 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 7, pp. 104–112
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 2 (Zero-Pair Tactile Texture) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE ZERO-PAIR EXTINCTION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Top-left: Place one positive x^2 flat ($+x^2$).

Step 2: Top-right: x meets (−2), producing two negative x-rods (−2x, ribbed).

Step 3: Bottom-left: (+2) meets x, producing two positive x-rods (+2x, smooth).

Step 4: Pair each positive x-rod with a negative x-rod. Notice: (+2x) + (−2x) = 0! Remove them from the board!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Annihilation Invariant: (+x) + (−x) = 0 -> Net Area = $x^2 - 4$

PHYSICAL APPARATUS MANIFEST:

1 x^2 flat, 2 positive x-rods (smooth), 2 negative x-rods (ribbed), 4 negative units

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE REMAINING CORNER

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Inspect the bottom-right corner where (+2) meets (−2).

FORM A MASTER STATION TASKS (Complete All 3):

1. **What tiles fill the corner? (+2) × (−2).**
   Expected Physical Finding: **Four negative unit tiles (−4, ribbed)**

2. **What tiles remain on the entire board after zero-pairs vanish?**
   Expected Physical Finding: **One positive x^2 flat and four negative units**

3. **State the algebraic result of (x + 2)(x − 2).**
   Expected Physical Finding: **$x^2 - 4$ (Difference of Squares!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x] [2] [+] [ENTER] [x] [2] [−] [*] -> x^2 − 4

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x+2	X: x + 2	Binomial sum
ENTER	Y: x + 2 | X: x + 2	Stack lift
x−2	Y: x + 2 | X: x − 2	Binomial difference
*	X: $x^2 - 4$	Difference of squares: cross terms annihilate

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Completing the Square

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did we have to divide the 6 rods into two equal halves of 3?'
- Ask: 'How does $(b/2)^2$ physically appear in the corner of the square?'
- Ask: 'How does this physical exercise turn into the quadratic formula $-b \pm \sqrt{b^2-4ac}/2a$?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students forgetting to add 9 to BOTH sides of the equation. Emphasize that adding physical tiles to the left r
equires balancing the right.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Complete the square for $x^2 + 10x$. How many unit tiles must be added to fill the corner hole?"
""

OFFICIAL SOLUTION KEY:

Split 10 into 5 and 5. Corner hole is $5 \times 5 = 25$ unit tiles.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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at the QR code to launch the interactive 3D model,
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Completing the Square

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is it called 'Completing the Square'? Because there is a literal, physical hole in the corner of your geometric square waiting to be filled.

STEP-BY-STEP APPARATUS SETUP:

1. Take 1 x^2 flat and 6 x-rods.
2. Your goal: Build a PERFECT geometric square.
3. Start by placing the x^2 flat at the corner.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: 1 x^2 flat, 6 x-rods, 9 unit tiles + SC-32 Hardware Brent Solver

DATUM TARGET: Square Completion: $x^2 + 6x + 9 = (x + 3)^2$

CALCULATOR KEY: [x^2 + 6x - 16 = 0] [SOLVE] -> x = 2, x = -8

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 2 (Completing the Square)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Geometric Square Completion)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 1, Ch. 9, Sec. 9.4 • CCSS.MATH.HSA.REI.B.4.B • TEKS A.8A
GRADE LEVEL BAND:	Grades 9–11 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 7, pp. 112–124
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 7 (RPN Solver Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: SYMMETRIC SPLITTING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: If you put all 6 x-rods on the right, you get an elongated rectangle, not a square.

Step 2: To keep the shape square, divide the 6 x-rods EQUALLY: 3 along the top and 3 along the side (b/2 = 6/2 = 3).

Step 3: Look at the bottom-right corner: There is an empty square hole of dimension 3 by 3!

Step 4: Fill the hole with unit tiles: Exactly 3 × 3 = 9 unit tiles are needed to COMPLETE the square! Result: $(x + 3)^2 = x^2 + 6x + 9$.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Square Completion: $x^2 + 6x + 9 = (x + 3)^2$

PHYSICAL APPARATUS MANIFEST:

1 x^2 flat, 6 x-rods, 9 unit tiles + SC-32 Hardware Brent Solver

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: SOLVING $x^2 + 6x = 16$

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Use your completed square to solve the equation $x^2 + 6x = 16$.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Add 9 to both sides: $(x^2 + 6x) + 9 = 16 + 9$.**
   Expected Physical Finding: $(x + 3)^2 = 25$

2. **Take square root of both sides: sqrt($(x + 3)^2$) = sqrt(25).**
   Expected Physical Finding: $x + 3 = \pm 5$

3. **Solve for x: x = −3 + 5 and x = −3 − 5.**
   Expected Physical Finding: x = 2 or x = −8

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x^2 + 6x − 16 = 0] [SOLVE] –> x = 2, x = −8

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
eqn	X: $x^2 + 6x - 16$	Define quadratic equation
SOLVE	X: 2 | Y: −8	Brent-Dekker root solver finds real roots via square completion

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: Instant Eyes: Subitizing to 5

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it easier to see 4 dots in a row than 4 dots scattered on the floor?'
- Ask: 'How does your brain use the full row of 5 to know what is missing?'
- Ask: 'What happens when you add 1 more to a full row of 5?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students reverting to touching and counting 1, 2, 3, 4. Encourage them to trust their eyes: 'Full row minus 1 is 4!'

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A frame has 3 tokens in the top row and 1 token in the bottom row. What is the total without counting by ones?"
""

OFFICIAL SOLUTION KEY:

3 in top + 1 in bottom = 4 tokens (or 5 minus 1).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: Instant Eyes: Subitizing to 5

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Can your eyes recognize a group of objects instantly without counting them 1, 2, 3? In this lab, train your brain to subitize using the 5-anchor grid.

STEP-BY-STEP APPARATUS SETUP:

1. Place the Ten-Frame Tray in front of you.
2. Notice the top row has exactly 5 circular sockets.
3. Pull 5 tokens (smooth shiny side facing up).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Ten-Frame Tray (2×5 Grid) + 10 Dual-Texture Optical Flip Tokens
DATUM TARGET:	Perceptual Invariant: 5-Frame Anchor Allows Subitizing Without One-by-One Counting
CALCULATOR KEY:	[1] [ENTER] [1] [+] -> 2 (Step Counter)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Kindergarten (KA), Chapter 1 (Numbers to 5)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Kindergarten, Unit 1 (Five-Frames & Counting)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Kindergarten, Ch. 1 • CCSS.MATH.K.CC.B.4 • TEKS K.2A
GRADE LEVEL BAND:	Grades K-1 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 3 ('Counting & Place Value'), pp. 38-52
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 2 (5-Anchor Geometry) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 5-FRAME FLASH GAME

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Place 3 tokens in the top row. Look away. Look back for 1 second. Notice: You instantly see 3!

Step 2: Place 5 tokens in the top row. Notice: A completely full top row ALWAYS means 5 without counting.

Step 3: Partner Flash: Partner A places 1 to 5 tokens, then covers the frame with their hand.

Step 4: Partner A reveals the frame for 1 second, then covers it. Partner B announces the number instantly!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Perceptual Invariant: 5-Frame Anchor Allows Subitizing Without One-by-One Counting

PHYSICAL APPARATUS MANIFEST:

Ten-Frame Tray (2×5 Grid) + 10 Dual-Texture Optical Flip Tok
ens

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: HOW MANY ARE MISSING?

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Look at the empty sockets in the 5-row to find the complement.

FORM A MASTER STATION TASKS (Complete All 3):

1. **If 4 tokens are placed in the top row, how many empty sockets remain?**
   Expected Physical Finding: 1 empty socket (4 + 1 = 5)

2. **If 2 tokens are placed, how many are empty?**
   Expected Physical Finding: 3 empty sockets (2 + 3 = 5)

3. **If 0 tokens are placed, how many are empty?**
   Expected Physical Finding: 5 empty sockets (0 + 5 = 5)

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [ENTER] [1] [+] -> 2 (Step Counter)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Key 1
ENTER	Y: 1 | X: 1	Stack lift
1	Y: 1 | X: 1	Step increment
+	X: 2	Step counter advance

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: Complements of Ten

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do 7 and 3 always belong together when making 10?'
- Ask: 'How does knowing 6 + 4 = 10 help you solve 60 + 40 later on?'
- Ask: 'Why is 5 + 5 the only double fact on the ten-frame?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students losing track of which side is which. The shiny side represents Addend A; the frosted side represents Addend B.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A student holds 10 coins. 7 are heads. How many are tails without counting them one by one?"
""

OFFICIAL SOLUTION KEY:

10 − 7 = 3 tails.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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https://stackcalc.io/products/ten-frames/#product-videos

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FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: Complements of Ten

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is 10 the most important number in our counting system? In this lab, flip two-texture tokens to discover all the 'Ten-Friends' that fill the frame.

STEP-BY-STEP APPARATUS SETUP:

1. Fill all 10 sockets of your Ten-Frame Tray with tokens.
2. Start with all 10 tokens showing the shiny specular side (10 + 0 = 10).
3. Prepare to flip tokens one by one to the frosted micro-grooved side.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Ten-Frame Tray + 10 Two-Sided Flip Tokens (Specular Smooth vs Micro-Grooved)
DATUM TARGET:	Conservation of Ten: A + B = 10 (Full 2×5 Frame is Invariant Landmark)
CALCULATOR KEY:	[7] [ENTER] [10] [−] -> 3 (Complement of 7)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 2 (Number Bonds to 10)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 1, Module 3 (Ten-Frames & Making Ten)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 1, Sec. 1.4 • CCSS.MATH.1.OA.C.6 • TEKS 1.3C
GRADE LEVEL BAND:	Grades K–2 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 3, pp. 52–64
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 5 (Two-Color Optical Contrast) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: FLIPPING THE TEN-FRIENDS

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Flip 1 token to frosted: 9 shiny + 1 frosted = 10.

Step 2: Flip another token: 8 shiny + 2 frosted = 10.

Step 3: Flip another token: 7 shiny + 3 frosted = 10.

Step 4: Continue down the line: 6+4, 5+5, 4+6, 3+7, 2+8, 1+9. Notice: The frame is ALWAYS completely full!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Conservation of Ten: A + B = 10 (Full 2×5 Frame is Invariant Landmark)

PHYSICAL APPARATUS MANIFEST:

Ten-Frame Tray + 10 Two-Sided Flip Tokens (Specular Smooth v
s Micro-Grooved)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: BLIND TEN-COMPLEMENT AUDIT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Close your eyes. Partner A flips a secret number of tokens to frosted.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Partner B opens eyes, sees 6 shiny tokens. How many are frosted?**
   Expected Physical Finding: **4 frosted tokens (6 + 4 = 10)**

2. **Partner B sees 2 shiny tokens. How many are frosted?**
   Expected Physical Finding: **8 frosted tokens (2 + 8 = 10)**

3. **Key [10] [ENTER] [6] [−] on SC-6.**
   Expected Physical Finding: **Register X: 4**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [7] [ENTER] [10] [−] -> 3 (Complement of 7)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
10	X: 10	Target frame capacity
ENTER	Y: 10 | X: 10	Stack lift
7	Y: 10 | X: 7	Addend 7
−	X: 3	Extract ten-complement 3

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: Bridging Ten Across the 5mm Rib

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it so much easier to add 10 + 3 than 8 + 5 in your head?'
- Ask: 'How does the 5mm raised divider rib stop your eyes from seeing 12 instead of 10 + 2?'
- Ask: 'Can you use this trick to solve 8 + 7 in 2 seconds?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to fill Frame 2 before Frame 1 is complete. Enforce the Golden Rule: 'Always complete the ten first!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Use bridging to solve 8 + 6. What two numbers do you break 6 into?"
""

OFFICIAL SOLUTION KEY:

Break 6 into 2 and 4. (8 + 2) + 4 = 10 + 4 = 14.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: Bridging Ten Across the 5mm Rib

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How does mental math work for 8 + 7 or 9 + 5? Stop counting on your fingers: bridge across the 5mm architectural divider to make ten first!

STEP-BY-STEP APPARATUS SETUP:

1. Snap two Ten-Frame Trays together end-to-end using the octagonal MultiBuild connectors.
2. Frame 1 is the 'Tens Anchor'; Frame 2 is the 'Overflow Stage'.
3. Place 8 shiny tokens in Frame 1. Hold 5 frosted tokens in your hand (8 + 5).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Two Ten-Frame Trays (Docked End-to-End via MultiBuild Snaps) + 20 Tokens

DATUM TARGET: Regrouping Invariant: Fill the First Frame to 10 First, Overflow to Second Frame

CALCULATOR KEY: [8] [ENTER] [5] [+] -> 13

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 1A, Chapter 7 (Addition with Regrouping)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 1, Unit 3 (Double Ten-Frames & Bridging)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 1, Ch. 4 • CCSS.MATH.1.OA.C.6 • TEKS 1.3D

GRADE LEVEL BAND: Grades 1–3 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 4 ('Operations & Regrouping'), pp. 68–82

HARDWARE UNBOXING BOOKLET: BASE-10 Booklet: Page 2 (5mm Architectural Rib) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE OVERFLOW LOCK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at Frame 1: How many empty sockets are needed to make 10? (Exactly 2 sockets).

Step 2: Take 2 frosted tokens from your hand and lock them into Frame 1. Frame 1 is now FULL (10!).

Step 3: How many frosted tokens remain in your hand? (3 tokens).

Step 4: Place the remaining 3 tokens into Frame 2 across the 5mm divider. Look at the total: 1 full ten + 3 units = 13!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Regrouping Invariant: Fill the First Frame to 10 First, Overflow to Second Frame

PHYSICAL APPARATUS MANIFEST:

Two Ten-Frame Trays (Docked End-to-End via MultiBuild Snaps)
+ 20 Tokens

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE 9-BRIDGE SHORTCUT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Solve 9 + 6 using the bridging strategy.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Place 9 tokens in Frame 1. How many are needed to make 10?**
   Expected Physical Finding: **Only 1 token**

2. **Take 1 from 6 to complete Frame 1. How many remain for Frame 2?**
   Expected Physical Finding: **5 tokens remain (6 – 1 = 5)**

3. **State the total immediately: 1 ten + 5 units.**
   Expected Physical Finding: **15 (9 + 6 = 15)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [8] [ENTER] [5] [+] -> 13

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
8	X: 8	First addend
ENTER	Y: 8 | X: 8	Stack lift
5	Y: 8 | X: 5	Second addend
+	X: 13	Combine and bridge into 1 ten and 3 units

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: The 10-for-1 Base-Ten Regrouping

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can you never have more than 9 loose unit cubes in your answer pile?'
- Ask: 'How does trading 10 cubes for 1 rod make counting faster?'
- Ask: 'What does the little carried 1 at the top of your paper represent in real plastic?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students setting aside 10 cubes without bringing a ten-rod into the total. Always enforce: 'Nothing leaves the bank without a trade!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"You have 3 ten-rods and 15 unit cubes. What number do you actually have after trading?"
""

OFFICIAL SOLUTION KEY:

15 units trades for 1 rod + 5 units. Total = 4 rods and 5 units = 45.

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SCAN FOR 3D SOLUTION

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BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: The 10-for-1 Base-Ten Regrouping

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does carrying a 1 to the next column work in vertical addition? In this lab, you physically trade 10 loose unit cubes for 1 solid ten-rod at the bank.

STEP-BY-STEP APPARATUS SETUP:

1. Place all loose unit cubes on the left ('Ones Bank').
2. Place all solid ten-rods in the center ('Tens Bank').
3. Problem: Add 25 + 18 using physical blocks.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Base-10 Blocks: 20 Unit Cubes (10×10×10mm), 10 Ten-Rods (10×10×100mm)

DATUM TARGET: Exchange Invariant: Exactly 10 Unit Cubes Snap/Align into 1 Ten-Rod (10U = 100mm)

CALCULATOR KEY: [25] [ENTER] [18] [+] -> 43

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2A, Chapter 1 (Numbers to 1,000)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 2 (Place Value & Regrouping)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 5 • CCSS.MATH.2.NBT.A.1 • TEKS 2.2B
GRADE LEVEL BAND:	Grades 1–3 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 3, pp. 64–78
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 2 (10mm Metric Invariant) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE BANK EXCHANGE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Model 25: Take 2 ten-rods and 5 unit cubes.

Step 2: Model 18: Take 1 ten-rod and 8 unit cubes.

Step 3: Combine all loose units: 5 units + 8 units = 13 unit cubes. Notice: You have MORE than 10 loose cubes!

Step 4: The 10-for-1 Trade: Count out 10 loose units and trade them at the bank for 1 solid ten-rod.

Step 5: Count final blocks: 2 + 1 + 1 (new) = 4 ten-rods and 3 remaining units = 43!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Exchange Invariant: Exactly 10 Unit Cubes Snap/Align into 1 Ten-Rod (10U = 100mm)

PHYSICAL APPARATUS MANIFEST:

Base-10 Blocks: 20 Unit Cubes (10×10×10mm), 10 Ten-Rods (10×10×100mm)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: CARRYING TWICE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 47 + 36. Track the unit trade.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Combine units: 7 + 6.**
   Expected Physical Finding: **13 units**
2. **Execute the trade: 10 units -> 1 ten-rod.**
   Expected Physical Finding: **1 new rod, 3 units left**
3. **Combine all rods: 4 + 3 + 1 (traded).**
   Expected Physical Finding: **8 rods**
4. **Final total.**
   Expected Physical Finding: **83**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [25] [ENTER] [18] [+] -> 43

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
25	X: 25	2 rods, 5 units
ENTER	Y: 25 | X: 25	Stack lift
18	Y: 25 | X: 18	1 rod, 8 units
+	X: 43	Regroup: 13 units becomes 1 new rod + 3 units (4 rods, 3 units)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: Decomposing the Ten: Subtraction

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did 52 stay equal to 52 even after we traded a rod for 10 cubes?'
- Ask: 'When you cross out the 5 and write 4 on paper, which block did you move?'
- Ask: 'When you write a little 1 next to the 2 to make 12, where did those 10 cubes come from?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students subtracting smaller from larger in reverse (e.g., doing 7 − 2 = 5 in 52 − 27). Remind them: you must take away from what you STARTED with!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Solve 63 − 28 using blocks. What does 63 decompose into?"
""

OFFICIAL SOLUTION KEY:

63 decomposes into 5 rods and 13 units. 13 − 8 = 5 units; 5 − 2 = 3 rods -> 35.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: Decomposing the Ten: Subtraction

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why do we cross out a number and write a smaller number when borrowing in subtraction? In this lab, you physically break open a ten-rod to get more units.

STEP-BY-STEP APPARATUS SETUP:

1. Problem: 52 − 27.
2. Model 52 on your desk: 5 ten-rods and 2 loose unit cubes.
3. Goal: Take away 27 (2 ten-rods and 7 unit cubes).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Base-10 Blocks: 5 Ten-Rods, 20 Unit Cubes + Regrouping Tray

DATUM TARGET: Decomposition Invariant: 1 Ten-Rod Unlocks into Exactly 10 Loose Unit Cubes

CALCULATOR KEY: [52] [ENTER] [27] [−] -> 25

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2A, Chapter 3 (Subtraction with Regrouping)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 3 (Subtraction Strategies)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 6 • CCSS.MATH.2.NBT.B.7 • TEKS 2.4C
GRADE LEVEL BAND:	Grades 2–4 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 82–96
HARDWARE UNBOXING BOOKLET:	BASE-10 Booklet: Page 6 (Aha! Discovery: Decomposition Proof)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE UNLOCKING STEP

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Try to take away 7 loose units from your pile: You only have 2 loose cubes! You are stuck!

Step 2: The Bank Trade: Pick up 1 of your ten-rods and trade it at the bank for 10 loose unit cubes.

Step 3: Add the 10 cubes to your 2 loose cubes: Now you have 4 ten-rods and 12 loose cubes (40 + 12 = 52).

Step 4: Now take away 7 cubes: 12 − 7 = 5 unit cubes remain.

Step 5: Now take away 2 ten-rods: 4 − 2 = 2 ten-rods remain. Total answer = 25!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Decomposition Invariant: 1 Ten-Rod Unlocks into Exactly 10 Loose Unit Cubes

PHYSICAL APPARATUS MANIFEST:

Base-10 Blocks: 5 Ten-Rods, 20 Unit Cubes + Regrouping Tray

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: SUBTRACTING ACROSS ZERO

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 40 − 16. How do you take 6 units from zero units?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Model 40: 4 ten-rods and 0 units.**
   Expected Physical Finding: **Cannot take 6 from 0**

2. **Trade 1 rod for 10 units: What blocks do you hold?**
   Expected Physical Finding: **3 rods and 10 units**

3. **Take away 16 (1 rod and 6 units). What remains?**
   Expected Physical Finding: **2 rods and 4 units = 24**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [52] [ENTER] [27] [−] -> 25

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
52	X: 52	Minuend: 5 rods, 2 units
ENTER	Y: 52 | X: 52	Stack lift
27	Y: 52 | X: 27	Subtrahend: 2 rods, 7 units
−	X: 25	Subtract: 1 rod decomposes into 10 units (12 − 7 = 5 units, 4 − 2 = 2 rods)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Hundreds, Flats & Metric Volume

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'How can the exact same plastic block be 100 on Monday and 0.1 on Friday?'
- Ask: 'Why does moving the decimal point right multiply by 10?'
- Ask: 'How does this connect our base-10 blocks to the Fraction Studio towers?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking decimals are a completely new kind of number. Emphasize: place value simply shifts whether t
he unit is small or large.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Model 2.35 using flats, rods, and units (where Flat = 1, Rod = 0.1, Unit = 0.01). What pieces do you pull?"
""

OFFICIAL SOLUTION KEY:

2 Flats (2.0), 3 Ten-Rods (0.3), and 5 Unit Cubes (0.05).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/ten-frames/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Hundreds, Flats & Metric Volume

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is our number system metric? A single unit cube is exactly 1 cubic centimeter. Ten of them make a rod, a hundred make a flat, and a thousand make a liter of

STEP-BY-STEP APPARATUS SETUP:

1. Line up 10 ten-rods side-by-side on your desk.
2. Place a Hundred-Flat on top of the 10 rods.
3. Notice: The flat matches the 10 rods in every dimension (100 × 100 mm).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Base-10 Blocks: 2 Hundred-Flats (10×100×100mm), 10 Ten-Rods, 10 Unit Cubes

DATUM TARGET: Metric Proportionality: 1 Cube = 1cm³, 1 Rod = 10cm³, 1 Flat = 100cm³, 1 Large Cube = 1,000cm³ (1 Liter)

CALCULATOR KEY: [345] [ENTER] [278] [+] -> 623

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 3A, Chapter 1 (Numbers to 10,000)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 3, Unit 1 (Place Value with 3-Digit Numbers)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 3, Ch. 1 • CCSS.MATH.3.NBT.A.2 • TEKS 3.2A

GRADE LEVEL BAND: Grades 3-5 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 3, pp. 78-88

HARDWARE UNBOXING BOOKLET: BASE-10 Booklet: Page 7 (RPN Register Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE HUNDREDS REGULATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: 10 units = 1 rod (10). 10 rods = 1 flat (100).

Step 2: Stack 10 Hundred-Flats on top of each other: It forms a perfect 1000-cube (100 × 100 × 100 mm).

Step 3: Measure: 1000-cube volume = 1 Liter = 1,000 grams of water (1 kilogram).

Step 4: Conclude: In base-10, every place value step is a 10× multiplication in physical space.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Metric Proportionality: 1 Cube = 1cm³, 1 Rod = 10cm³, 1 Flat = 100cm³, 1 Large Cube = 1,000cm³ (1 Liter)

PHYSICAL APPARATUS MANIFEST:

Base-10 Blocks: 2 Hundred-Flats (10×100×100mm), 10 Ten-Rods,
10 Unit Cubes

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

BASE-10 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: DECIMAL REINTERPRETATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Declare the large 1000-cube to be '1 WHOLE'.

FORM A MASTER STATION TASKS (Complete All 3):

1. **If the 1000-cube is 1 WHOLE, what does a Hundred-Flat represent?**
   Expected Physical Finding: **0.1 (One Tenth)**

2. **What does a Ten-Rod represent?**
   Expected Physical Finding: **0.01 (One Hundredth)**

3. **What does a single Unit Cube represent?**
   Expected Physical Finding: **0.001 (One Thousandth)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [345] [ENTER] [278] [+] –> 623

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
345	X: 345	3 flats, 4 rods, 5 units
ENTER	Y: 345 | X: 345	Stack lift
278	Y: 345 | X: 278	2 flats, 7 rods, 8 units
+	X: 623	Regroup both columns: 13 units –> 1 rod, 12 rods –> 1 flat (6 flats, 2 rods, 3 units)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: Double Trouble: Subitizing Doubles

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does tile 6|6 fill the 12-frame tray completely?'
- Ask: 'How does knowing 4 + 4 = 8 help you solve 4 + 5?'
- Ask: 'Why do domino pips make patterns of 5 look like an X and 6 look like parallel lines?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students counting pips one-by-one. Encourage instant recognition: 'I see 5 and 5, so it must be 10!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A mystery double tile has 8 pips in total. What tile is it?"
""

OFFICIAL SOLUTION KEY:

Tile 4|4 (since 4 + 4 = 8).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: Double Trouble: Subitizing Doubles

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why are doubles facts (4+4, 5+5, 6+6) the fastest facts in mathematics? Because your brain loves mirror symmetry. In this lab, find all 6 doubles tiles.

STEP-BY-STEP APPARATUS SETUP:

1. Spread all 28 dominoes face up on the table.
2. Find all the 'Double Tiles' where both ends match: 1|1, 2|2, 3|3, 4|4, 5|5, 6|6.
3. Line them up in ascending order.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: 28 Double-Six Dominos + 12-Frame Regrouping Tray

DATUM TARGET: Doubles Invariant: 2 × N is Always Even, Instantly Recognizable by Parallel Symmetry

CALCULATOR KEY: [3] [ENTER] [2] [*] -> 6 (Double 3)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 4 (Doubles & Addition)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 2 (Doubles & Domino Arrays)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 2 • CCSS.MATH.1.OA.C.6 • TEKS 1.3D
GRADE LEVEL BAND:	Grades K-2 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4 ('Operations & Skip-Counting'), pp. 72-80
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 2 (Combinatoric Distribution) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE SYMMETRY RACE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Point to 1|1: 1 + 1 = 2.

Step 2: Point to 2|2: 2 + 2 = 4. Point to 3|3: 3 + 3 = 6.

Step 3: Point to 4|4: 4 + 4 = 8. Point to 5|5: 5 + 5 = 10. Point to 6|6: 6 + 6 = 12.

Step 4: Notice: Every single double total is an EVEN number (2, 4, 6, 8, 10, 12).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Doubles Invariant: 2 × N is Always Even, Instantly Recognizable by Parallel Symmetry

PHYSICAL APPARATUS MANIFEST:

28 Double-Six Dominos + 12-Frame Regrouping Tray

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: BLIND GRAB DOUBLES DRILL

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Partner A flashes a double tile for 1 second. Partner B calls out the sum.

FORM A MASTER STATION TASKS (Complete All 3):

1 **Flash tile [4|4]. What is the sum?**
Expected Physical Finding: 8

2 **Flash tile [6|6]. What is the sum?**
Expected Physical Finding: 12 (Fills the 12-frame tray exactly!)

3 **Can an odd number ever be a double fact?**
Expected Physical Finding: Never! A double is always two equal halves (even).

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [ENTER] [2] [*] -> 6 (Double 3)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3	X: 3	Base operand 3
ENTER	Y: 3 | X: 3	Stack lift
2	Y: 3 | X: 2	Scale factor 2
*	X: 6	Multiply by 2 evaluates double

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: Near-Doubles & Compensation

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it faster to do 6 + 6 + 1 than counting 7 dots after 6?'
- Ask: 'Are near-doubles sums always odd or always even?'
- Ask: 'How does this trick turn 12 hard math facts into instant mental recall?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding 1 to the wrong number. Emphasize: 'Double the smaller number and add 1, OR double the larger a
nd subtract 1.'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Use near-doubles to solve 7 + 8 in your head. What is your thought process?"
""

OFFICIAL SOLUTION KEY:

Double 7 is 14. 14 + 1 = 15 (or double 8 is 16; 16 − 1 = 15).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: Near-Doubles & Compensation

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What do you do if you forget what 5 + 6 is? If you know 5 + 5 = 10, then 5 + 6 is just 'one more'!

STEP-BY-STEP APPARATUS SETUP:

1. Pull tile 3|3 and tile 3|4 from your domino tray.
2. Place them side-by-side.
3. Compare their pip arrays.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Dominoes: Doubles 3|3, 4|4, 5|5 and Near-Doubles 3|4, 4|5, 5|6
DATUM TARGET:	Compensation Rule: N + (N + 1) = 2N + 1 (Doubles Plus One)
CALCULATOR KEY:	[4] [ENTER] [2] [*] [1] [+] -> 9 (4 + 5 = 2×4 + 1)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 4, Lesson 4.3 (Doubles Plus One)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 2, Module 2 (Near-Doubles Strategy)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 2, Sec. 2.4 • CCSS.MATH.1.OA.C.6 • TEKS 1.3D
GRADE LEVEL BAND:	Grades 1–2 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 80–86
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 5 (Tutorial 1) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE PLUS-ONE EXTRA PIP

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at 3|3: exactly 3 on the left, 3 on the right = 6.

Step 2: Look at 3|4: it has 3 on the left, but 4 on the right. 4 is just 3 + 1!

Step 3: So 3|4 is just double 3 plus 1: (3 + 3) + 1 = 6 + 1 = 7.

Step 4: Now test tile 4|5: Double 4 is 8, plus 1 is 9! (4 + 5 = 9).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Compensation Rule: N + (N + 1) = 2N + 1 (Doubles Plus One)

PHYSICAL APPARATUS MANIFEST:

Dominoes: Doubles 3|3, 4|4, 5|5 and Near-Doubles 3|4, 4|5, 5|6

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: DOUBLES MINUS ONE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you also solve 4 + 5 using Doubles MINUS One? (5 + 5 − 1).

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is 5 + 5?**
   Expected Physical Finding: **10**

2. **Subtract 1: 10 − 1 = ?**
   Expected Physical Finding: **9**

3. **Does Doubles Plus 1 (8 + 1) match Doubles Minus 1 (10 − 1)?**
   Expected Physical Finding: **Both equal 9! Two paths to the same truth.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [4] [ENTER] [2] [*] [1] [+] –> 9 (4 + 5 = 2×4 + 1)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
4	X: 4	Base double 4
2	Y: 4 | X: 2	Double factor
*	X: 8	Double total = 8
1	Y: 8 | X: 1	Plus one compensation
+	X: 9	Final near-double result = 9

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: The 12-Frame Associative Train

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why are there so many ways to make 12?'
- Ask: 'How does the associative property let you rearrange the tiles in any order?'
- Ask: 'What happens if you try to make 12 using three identical tiles?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students counting blank ends as 1. Remind them: blank means ZERO pips!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Can you make 12 using three dominoes where every domino is a double tile?"
""

OFFICIAL SOLUTION KEY:

Yes: [1|1] (2) + [2|2] (4) + [3|3] (6) = 2 + 4 + 6 = 12!

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: The 12-Frame Associative Train

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does the domino tray have exactly 12 sockets? Because 12 is the ultimate composite number, divisible by 1, 2, 3, 4, 6, and 12.

STEP-BY-STEP APPARATUS SETUP:

1. Dock the 12-Frame Regrouping Tray in the center of the table.
2. Notice: Tile [6|6] has exactly 12 pips and fills the entire tray capacity.
3. Challenge: Can you find TWO different dominoes that add up to exactly 12 pips?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	12-Frame Tray + Dominoes [6|6], [5|5], [4|6], [3|6], [2|5], etc.
DATUM TARGET:	Associative Invariant: (A + B) + C = A + (B + C) = 12 (Full Tray Capacity)
CALCULATOR KEY:	[4] [ENTER] [5] [+] [3] [+] -> 12

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1B, Chapter 10 (Addition to 20)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 1, Module 2 (Combinations to 12)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 3 • CCSS.MATH.1.OA.B.3 • TEKS 1.3D
GRADE LEVEL BAND:	Grades 1-3 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 86-92
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 2 (12-Frame Tray Geometry) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: DOCKING DOMINO PAIRS

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Pull tile [4|3] (7 pips). What second tile is needed to make 12? (12 − 7 = 5 pips).

Step 2: Find a tile with 5 pips (e.g. [2|3] or [1|4] or [0|5]). Dock them both in the tray.

Step 3: Count total pips: 7 + 5 = 12! The pair balances the tray.

Step 4: Now find THREE dominoes that add to 12: e.g. [1|2] (3) + [2|2] (4) + [1|4] (5) = 12.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Associative Invariant: (A + B) + C = A + (B + C) = 12 (Full Tray Capacity)

PHYSICAL APPARATUS MANIFEST:

12-Frame Tray + Dominoes [6|6], [5|5], [4|6], [3|6], [2|5], etc.

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: ALL-DOMINO 12-TRAIN

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

How many unique pairs of dominoes in the box total 12 pips?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Pair 1: Tile [6|5] (11 pips) + ?**
   Expected Physical Finding: **Tile [0|1] or [1|0] (1 pip)**

2. **Pair 2: Tile [5|5] (10 pips) + ?**
   Expected Physical Finding: **Tile [1|1] or [0|2] (2 pips)**

3. **Pair 3: Tile [4|5] (9 pips) + ?**
   Expected Physical Finding: **Tile [0|3] or [1|2] (3 pips)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [4] [ENTER] [5] [+] [3] [+] -> 12

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
4	X: 4	First tile pip value
5	Y: 4 | X: 5	Second pip value
+	X: 9	Subtotal = 9
3	Y: 9 | X: 3	Third pip value
+	X: 12	Fills the 12-frame tray exactly

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 4: The 28-Tile Triangular Pyramid

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did row lengths shrink by exactly 1 in each step?'
- Ask: 'Why didn\'t we include tile [3|1] in the 3s row?' (Already included as [1|3]!).
- Ask: 'How does the pyramid prove that no dominoes were duplicated or forgotten?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to count [1|2] and [2|1] as two separate tiles. Explain that dominoes are unordered pairs—rota
ting the tile doesn't make a new piece!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"What is the total sum of all pips across the entire 28-tile set?"
""

OFFICIAL SOLUTION KEY:

Every number 0–6 appears 8 times. Total = 8 × (0+1+2+3+4+5+6) = 8 × 21 = 168 pips.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

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- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 1: The 28-Tile Triangular Pyramid

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does a domino set have exactly 28 tiles instead of 25 or 30? Because 28 is a triangular number formed by all possible pair combinations of numbers 0 through

STEP-BY-STEP APPARATUS SETUP:

1. Clear your table.
2. Your goal: Build the grand 7-row triangular domino pyramid.
3. Row 0 (Blanks): 7 tiles with a blank end: [0|0], [0|1], [0|2], [0|3], [0|4], [0|5], [0|6].

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Complete Set of 28 Double-Six Dominos + Table Workspace

DATUM TARGET: Combinatoric Invariant: Pairs from {0,1,2,3,4,5,6} with replacement = 7×8/2 = 28 Tiles

CALCULATOR KEY: [7] [ENTER] [8] [*] [2] [/] -> 28 (T_7 formula n(n+1)/2)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3B, Chapter 14 (Patterns & Combinations)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 4, Unit 6 (Patterns, Arrays & Combinatorics)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 4, Ch. 10 • CCSS.MATH.4.OA.C.5 • TEKS 4.5B
GRADE LEVEL BAND:	Grades 3–5 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7 ('Data & Probability'), pp. 162–170
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 2 (T_7 Triangular Distribution) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 2: LAB INVESTIGATION: BUILDING THE TRIANGLE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Lay the 7 blank tiles in a long horizontal row at the bottom (Row 1: 7 tiles).

Step 2: Above it, lay all tiles starting with 1 (excluding [1|0] already used): [1|1], [1|2], [1|3], [1|4], [1|5], [1|6] (Row 2: 6 tiles).

Step 3: Row 3: All remaining tiles starting with 2: [2|2], [2|3], [2|4], [2|5], [2|6] (5 tiles).

Step 4: Continue upward: 4 tiles, 3 tiles, 2 tiles, and peak tile [6|6] (1 tile). Notice: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28 tiles!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Combinatoric Invariant: Pairs from {0,1,2,3,4,5,6} with replacement = 7×8/2 = 28 Tiles

PHYSICAL APPARATUS MANIFEST:

Complete Set of 28 Double-Six Dominos + Table Workspace

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE COMBINATORIC FORMULA

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Use the triangular formula T_n = n(n+1)/2 on your calculator.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Calculate T_7: 7 × 8 / 2.**
   Expected Physical Finding: **28 tiles (Double-Six Set)**
2. **How many tiles would a Double-Nine domino set have? (n = 10 elements: 0 to 9).**
   Expected Physical Finding: **10 × 11 / 2 = 55 tiles**
3. **How many tiles in a Double-Twelve set? (n = 13).**
   Expected Physical Finding: **13 × 14 / 2 = 91 tiles**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [7] [ENTER] [8] [*] [2] [/] -> 28 (T_7 formula n(n+1)/2)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
7	X: 7	Elements {0,1,2,3,4,5,6}
8	Y: 7 | X: 8	n + 1
*	X: 56	Permutation product
2	Y: 56 | X: 2	Symmetry divisor
/	X: 28	7th Triangular Number T_7 = 28 tiles

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 4: Turnaround Facts & Commutativity

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does rotation prove addition is commutative but not subtraction?'
- Ask: 'How does knowing turnaround facts cut your memorization list in half?'
- Ask: 'Does multiplication work like addition when you rotate a tile array?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking 2 − 5 is 3. Point out that having 2 apples and eating 5 is impossible in real physical space without owing 3.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A student knows 3 + 8 = 11. What other addition fact do they know for free?"
""

OFFICIAL SOLUTION KEY:

8 + 3 = 11 (and fact family subtractions 11 − 3 = 8 and 11 − 8 = 3).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 1: Turnaround Facts & Commutativity

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does 2 + 5 equal 5 + 2? You don't need a math degree to prove it—just turn the domino upside down in your hand.

STEP-BY-STEP APPARATUS SETUP:

1. Pick up tile [2|5] with your left hand.
2. Look at it: 2 pips on the left, 5 pips on the right (2 + 5 = 7).
3. Rotate the tile 180° in your fingers.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Dominoes [2|5], [3|4], [1|6], [2|4] + Stacking Desk

DATUM TARGET: Geometric Invariant: 180° Spatial Inversion Preserves Total Pip Mass (A + B = B + A)

CALCULATOR KEY: [2] [ENTER] [5] [+] [5] [ENTER] [2] [+] [−] -> 0

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 3 (Commutative Property of Addition)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 1, Module 4 (Turnaround Facts)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 3, Sec. 3.2 • CCSS.MATH.1.OA.B.3 • TEKS 1.3E
GRADE LEVEL BAND:	Grades 1–3 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 92–98
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 6 (Aha! Discovery: Rotational Commutativity)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 2: LAB INVESTIGATION: THE 180° ROTATION PROOF

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at the rotated tile: now 5 pips are on the left, and 2 are on the right (5 + 2 = 7).

Step 2: Did any pips appear out of thin air? Did any pips vanish? (No!).

Step 3: Conclude: The physical object is 100% identical. Therefore, 2 + 5 MUST equal 5 + 2.

Step 4: This is the Commutative Property of Addition (Turnaround Fact).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Geometric Invariant: 180° Spatial Inversion Preserves Total Pip Mass (A + B = B + A)

PHYSICAL APPARATUS MANIFEST:

Dominoes [2|5], [3|4], [1|6], [2|4] + Stacking Desk

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: DOES COMMUTATIVITY WORK FOR SUBTRACTION?

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Test 5 − 2 vs 2 − 5 using the same domino.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Calculate 5 − 2.**
   Expected Physical Finding: 3
2. **Try to take 5 pips away from 2 pips.**
   Expected Physical Finding: Cannot do it with physical pips! (Yields −3, not 3).
3. **Is subtraction commutative?**
   Expected Physical Finding: NO! Order matters for subtraction.

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [ENTER] [5] [+] [5] [ENTER] [2] [+] [−] –> 0

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
2	X: 2	Operand A
5	Y: 2 | X: 5	Operand B
+	X: 7	Forward sum = 7
5	Y: 7 | X: 5	Reverse operand B
2	Z: 7 | Y: 5 | X: 2	Reverse operand A
−	[illegible]	[illegible]

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 4: The Difference Race: Subtractive Fluency

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do all double tiles have a difference of zero?'
- Ask: 'Which tile gives the highest possible difference score?' ([6|0] = 6 points).
- Ask: 'How does finding the difference between two ends train your brain for division remainders?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding instead of subtracting. Emphasize: 'Difference means: how many MORE pips are on the big side t
han the small side?'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Two dominoes are drawn: Tile A is [5|1], Tile B is [6|3]. Which tile has the greater difference?"
""

OFFICIAL SOLUTION KEY:

Tile A difference = 5 − 1 = 4. Tile B difference = 6 − 3 = 3. Tile A wins!

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

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- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 1: The Difference Race: Subtractive Fluency

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Who has the fastest mental subtraction in the class? In this game, flip dominoes and subtract the smaller side from the larger side before your partner.

STEP-BY-STEP APPARATUS SETUP:

1. Shuffle all 28 dominoes face down in a draw pile ('The Boneyard').
2. Each player draws 5 dominoes and keeps them hidden.
3. Rule: The score of any domino is the DIFFERENCE between its two ends: High − Low.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Complete 28-Tile Domino Set + Timer / SC-6 Calculator

DATUM TARGET: Difference Invariant: Difference = High Pip Count − Low Pip Count (Non-Negative Range 0–6)

CALCULATOR KEY: [6] [ENTER] [2] [−] -> 4 (Tile 6|2 Difference)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2A, Chapter 2 (Mental Subtraction Strategies)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 2 (Difference as Distance on a Number Line)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 4 • CCSS.MATH.2.OA.B.2 • TEKS 2.4A
GRADE LEVEL BAND:	Grades 1–4 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 98–104
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 7 (RPN Postfix Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: SCORING DOMINO DIFFERENCES

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Inspect tile [6|1]: 6 − 1 = Difference of 5 points.

Step 2: Inspect tile [4|4]: 4 − 4 = Difference of 0 points (Double tiles score zero!).

Step 3: Inspect tile [5|2]: 5 − 2 = Difference of 3 points.

Step 4: Notice: The maximum possible difference in a Double-Six set is 6 (Tile [6|0]).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Difference Invariant: Difference = High Pip Count − Low Pip Count (Non-Negative Range 0–6)

PHYSICAL APPARATUS MANIFEST:

Complete 28-Tile Domino Set + Timer / SC-6 Calculator

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE TARGET DIFFERENCE RACE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Find all dominoes in the set that have a difference of exactly 2.

FORM A MASTER STATION TASKS (Complete All 3):

1. **List all difference-2 tiles.**
   Expected Physical Finding: [2|0], [3|1], [4|2], [5|3], [6|4]
2. **How many difference-2 tiles exist?**
   Expected Physical Finding: Exactly 5 tiles
3. **How many difference-0 tiles (doubles) exist?**
   Expected Physical Finding: Exactly 7 tiles ([0|0] through [6|6])

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [6] [ENTER] [2] [−] -> 4 (Tile 6|2 Difference)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
6	X: 6	High pip side
ENTER	Y: 6 | X: 6	Stack lift
2	Y: 6 | X: 2	Low pip side
−	X: 4	Subtract evaluates difference between ends

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Whole Truth

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is a 1/12 block shorter than a 1/3 block if 12 is greater than 3?'
- Ask: 'How does the whole datum frame act as an automatic error checker?'
- Ask: 'Can five 1/4 blocks fit inside the 84.00 mm whole frame? Why not?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 1/4 is bigger than 1/2 because 4 > 2. Direct them to stand the physical blocks side-by-side.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a tower made of three DIFFERENT unit fraction blocks. My summit levels exactly to the whole. None of my blocks"
" is 1/2. What blocks am I?"

OFFICIAL SOLUTION KEY:

Blocks [1/3], [1/4], and [1/12] (since 1/3 + 1/4 + 1/12 = 4/12 + 3/12 + 1/12 = 8/12... actually 1/2+1/3+1/6=1).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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https://stackcalc.io/products/fraction-studio/#product-videos

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Whole Truth

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does a bigger denominator mean a smaller piece? In this lab, you test the physical whole datum and discover inverse proportionality.

STEP-BY-STEP APPARATUS SETUP:

1. Place the Whole Frame [1/1] (84.00 mm tall) vertically on your desk.
2. Stack two [1/2] blocks. Compare their height against the [1/1] whole.
3. Stack four [1/4] blocks beside it. Observe the three summits.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks [1/1], [1/2], [1/3], [1/4], [1/6], [1/12] + Stacking Base

DATUM TARGET: H(1/n) = 84.00 / n mm -> H(1/1)=84.00, H(1/2)=42.00, H(1/12)=7.00 mm

CALCULATOR KEY: [1] [ENTER] [2] [/] -> 0.5

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3A, Chapter 6 (Understanding Fractions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 3, Unit 4 (Equal Shares & Wholes)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 3, Ch. 8 • CCSS.MATH.3.NF.A.1 • TEKS 3.3A
GRADE LEVEL BAND:	Grades 3–5 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5 ('Fractions & Ratios'), pp. 98–118
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 2 (Inverse Linear Formula) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE SUMMIT LEVELING CHECK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Lay a straightedge across Tower [1/1], Tower [1/2]+[1/2], and Tower four [1/4]s. All level to exactly 84.00 mm!

Step 2: Pull Block [1/12] (height 7.00 mm). How many [1/12] blocks make a half? (Stack 6 blocks).

Step 3: Try to make a whole with three [1/3] blocks. Confirm they match 84.00 mm.

Step 4: Notice: As denominator n grows (2 -> 3 -> 4 -> 12), block height shrinks (42 -> 28 -> 21 -> 7 mm).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
H(1/n) = 84.00 / n mm -> H(1/1)=84.00, H(1/2)=42.00, H(1/12)=7.00 mm

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks [1/1], [1/2], [1/3], [1/4], [1/6], [1/12] +
Stacking Base

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE MISSING PIECE PUZZLE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

If you stack one [1/2] block and one [1/3] block, what single block is needed to complete the 84.00 mm whole?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Measure 1/2 + 1/3 gap against the 84.00 mm frame.**
   Expected Physical Finding: **Gap = 14.00 mm**

2. **Which block has height 14.00 mm? Test [1/6].**
   Expected Physical Finding: **Block [1/6] levels the summit!**

3. **Write the fraction equation: 1/2 + 1/3 + ? = 1.**
   Expected Physical Finding: **1/2 + 1/3 + 1/6 = 1**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [ENTER] [2] [/] -> 0.5

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Numerator 1
ENTER	Y: 1 | X: 1	Stack lift
2	Y: 1 | X: 2	Denominator 2
/	X: 0.5	Divide: evaluates fraction value

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Multi-Face Transformer

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does 1/4 turn into 90° when talking about a clock or circle?'
- Ask: 'If you know 1/5 is 20%, what must 2/5 and 3/5 be?'
- Ask: 'How does rotating the block help when solving word problems with money?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 0.25 is larger than 0.5 because 25 > 5. Show them that two 0.25 blocks stacked equal one 0.5 0 block.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"My front face says 1/8. What does my decimal face, percent face, and angle face say?"
""

OFFICIAL SOLUTION KEY:

Decimal = 0.125, Percent = 12.5%, Angle = 45°.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/fraction-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Multi-Face Transformer

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Fractions, decimals, percentages, and circle angles are not four different subjects—they are just four faces of the exact same physical block!

STEP-BY-STEP APPARATUS SETUP:

1. Pick up Block [1/4] in your hand.
2. Look at the Front Face: notice debossed '1/4'.
3. Rotate the block 90° to the right, then 180°, then 270°.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks [1/2], [1/4], [1/5], [1/10] + Desk Comparator Stage

DATUM TARGET: Front: 1/4 | Right: 0.25 | Back: 25% | Left: 90° (Geometric Equivalence)

CALCULATOR KEY: [1] [/] [4] [F-D] -> 0.25

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 4B, Chapter 9 (Fractions & Decimals)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 4, Unit 3 (Decimals & Hundredths)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 4, Ch. 7 • CCSS.MATH.4.NF.C.6 • TEKS 4.2G
GRADE LEVEL BAND:	Grades 4–6 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 102–108
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 2 (4-Face Invariant) & Page 5 (Tutorial 1)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 4-WAY ROTATION AUDIT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Read the 4 faces of Block [1/4]: Front='1/4', Right='0.25', Back='25%', Left='90°'.

Step 2: Now inspect Block [1/2]: Front='1/2', Right='0.50', Back='50%', Left='180°'.

Step 3: Stack two [1/4] blocks with angle faces showing: 90° + 90° = 180°! It matches Block [1/2].

Step 4: Turn both blocks to the percent face: 25% + 25% = 50%! The physical reality never changes.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Front: 1/4 | Right: 0.25 | Back: 25% | Left: 90° (Geometric Equivalence)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks [1/2], [1/4], [1/5], [1/10] + Desk Comparator Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 4–6

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE MULTI-NOTATION EQUATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you build a tower that shows '1/2' + '0.25' + '25%' and prove it totals 100% and 360°?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Stack [1/2] (Front) + [1/4] (Right: 0.25) + [1/4] (Back: 25%).**
   Expected Physical Finding: **Total height = 84.00 mm (1 Whole)**

2. **Sum their angles: 180° + 90° + 90°.**
   Expected Physical Finding: **360° (Full Circle)**

3. **Convert 3/4 to decimal on SC-6: [3] [ENTER] [4] [/].**
   Expected Physical Finding: **0.75**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [/] [4] [F-D] -> 0.25

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Numerator
4	Y: 1 | X: 4	Denominator
/	X: 0.25	Decimal representation
F-D	X: 1/4	Toggle fraction / decimal display

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 4–6

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 4: Adding Like Fractions

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why didn't the size of the 1/8 blocks change when we stacked them?'
- Ask: 'What does the numerator tell you to do with your hands?'
- Ask: 'If you have 6/8 of a pizza, what simpler block matches that exact amount?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students writing 1/8 + 3/8 = 4/16. Hand them a 1/16 block and ask if their tower shrank!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a tower of like fractions with denominator 12. I level exactly with Block [3/4]. How many 1/12 blocks am I?"
""

OFFICIAL SOLUTION KEY:

Nine [1/12] blocks (9/12 = 3/4).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 1: Adding Like Fractions

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why don't we add the denominators when adding fractions? Building towers of eighths gives you an instant physical proof that the denominator never changes.

STEP-BY-STEP APPARATUS SETUP:

1. Pull four [1/8] blocks from your kit.
2. Stack three [1/8] blocks together (Tower 3/8).
3. Add one more [1/8] block to your tower (Tower 4/8).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks eight [1/8]s, four [1/4]s, two [1/2]s + Datum Stage

DATUM TARGET: 1/8 (10.5mm) × 4 = 42.00 mm = H(1/2)

CALCULATOR KEY: [1] [.] [8] [ENTER] [3] [.] [8] [+] -> 4/8 = 1/2

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3B, Chapter 11 (Fraction Addition)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 4, Unit 2 (Addition of Fractions)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 4, Ch. 7 • CCSS.MATH.4.NF.B.3.A • TEKS 4.3E
GRADE LEVEL BAND:	Grades 3–5 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 108–114
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 5 (Tutorial 1) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 2: LAB INVESTIGATION: THE EIGHTHS TOWER

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Stand your four [1/8] blocks beside Block [1/2]. Notice they level perfectly at 42.00 mm.

Step 2: If we added denominators, 1/8 + 3/8 would equal 4/16 (2.6 mm tall). Show that 4/16 would be tiny!

Step 3: Conclude: The denominator names the SIZE of the block (10.5 mm). The numerator counts HOW MANY blocks you hold.

Step 4: Test: Stack two [1/8] blocks. Verify they level with Block [1/4].

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
1/8 (10.5mm) × 4 = 42.00 mm = H(1/2)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks eight [1/8]s, four [1/4]s, two [1/2]s + Datum Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 3–5

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: BENCHMARK LEVELING

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

How many eighths must you add to 5/8 to reach the 84.00 mm whole summit?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Build Tower 5/8 (five [1/8] blocks). Measure remaining gap to whole.**
   Expected Physical Finding: Gap = 31.50 mm
2. **How many [1/8] blocks fit in 31.50 mm?**
   Expected Physical Finding: Three [1/8] blocks (3/8)
3. **Verify: 5/8 + 3/8 = 8/8 = 1 Whole.**
   Expected Physical Finding: Summit levels to 84.00 mm!

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [.] [8] [ENTER] [3] [.] [8] [+] -> 4/8 = 1/2

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1.8	X: 1/8	Key fraction 1/8
ENTER	Y: 1/8 | X: 1/8	Stack lift
3.8	Y: 1/8 | X: 3/8	Key fraction 3/8
+	X: 4/8	Add: 4/8 simplifies to 1/2 (0.5)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 3–5

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: Unlike Denominators

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did 12 work as a common denominator for both 3 and 4?'
- Ask: 'What is the connection between this lab and the Prime Studio LCM lab?'
- Ask: 'Could we have used 24ths to solve this problem? Why is 12 simpler?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding numerators and denominators (1/3 + 1/4 = 2/7). Show them that 2/7 would be shorter than 1/3 al
one!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Solve 2/3 + 1/6 using blocks. What is the sum, and does it simplify to a landmark block?"
""

OFFICIAL SOLUTION KEY:

2/3 = 4/6. 4/6 + 1/6 = 5/6 (Height = 70.00 mm).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/fraction-studio/#product-videos

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- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: Unlike Denominators

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

You cannot snap thirds and fourths together and read the answer on the blocks—unless you translate them into a common language: twelfths.

STEP-BY-STEP APPARATUS SETUP:

1. Stack one Block [1/3] on top of one Block [1/4] in the left comparator well.
2. Notice: The tower summit sits at 49.00 mm. No single block in your kit says '1/7' or '2/7'.
3. Challenge: How can we measure this tower using only identical pieces?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks [1/3], [1/4], and seven [1/12]s + Twin Comparator Well

DATUM TARGET: H(1/3) + H(1/4) = 28.00 + 21.00 = 49.00 mm = 7 × H(1/12)

CALCULATOR KEY: [1] [.] [3] [ENTER] [1] [.] [4] [+] -> 7/12

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 5A, Chapter 3 (Adding Unlike Fractions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 2 (Fraction Equivalence & Operations)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 5, Ch. 2 • CCSS.MATH.5.NF.A.1 • TEKS 5.3H
GRADE LEVEL BAND:	Grades 5–6 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 114–120
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 2 (Leveling Datum) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE TWELFTHS TRANSLATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Replace Block [1/3] with [1/12] blocks. Notice: Exactly four [1/12] blocks level to 1/3 (4/12).

Step 2: Replace Block [1/4] with [1/12] blocks. Notice: Exactly three [1/12] blocks level to 1/4 (3/12).

Step 3: Combine both twelfths piles: 4 twelfths + 3 twelfths = 7 twelfths (7/12).

Step 4: Stand your 7/12 tower beside the [1/3]+[1/4] tower. Confirm: Both summits level perfectly at 49.00 mm!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
H(1/3) + H(1/4) = 28.00 + 21.00 = 49.00 mm = 7 × H(1/12)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks [1/3], [1/4], and seven [1/12]s + Twin Compa
rator Well

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: SUBTRACTION WITH UNLIKE BLOCKS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 1/2 − 1/3 physically. Place Block [1/2] and Block [1/3] side-by-side in the comparator stage.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Measure the height difference between 1/2 (42.0mm) and 1/3 (28.0mm).**
   Expected Physical Finding: **Difference = 14.00 mm**

2. **Which block fits the difference exactly?**
   Expected Physical Finding: **Block [1/6] (14.00 mm) or two [1/12]s**

3. **Write the equation: 1/2 − 1/3 = ?**
   Expected Physical Finding: **3/6 − 2/6 = 1/6**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [.] [3] [ENTER] [1] [.] [4] [+] –> 7/12

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1.3	X: 1/3	Key fraction 1/3
ENTER	Y: 1/3 | X: 1/3	Stack lift
1.4	Y: 1/3 | X: 1/4	Key fraction 1/4
+	X: 7/12	Evaluate sum with common denominator 12

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 4: Improper Fractions & Mixed Numbers

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is 4/4 called a whole while 5/4 is called improper?'
- Ask: 'How does the 84.00 mm frame show the whole-number part of a mixed number?'
- Ask: 'Can you have a fraction greater than 2? What would 9/4 look like?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 5/4 is an invalid number. Explain that 'improper' is just a traditional name for a fraction greater than 1.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am an improper fraction with denominator 6. I am equal to 2 1/2. How many 1/6 blocks make me up?"
""

OFFICIAL SOLUTION KEY:

2 Wholes = 12/6. 1/2 = 3/6. Total = 15/6 (fifteen [1/6] blocks).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/fraction-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 1: Improper Fractions & Mixed Numbers

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What happens when you have more slices than fit in a single pie? In this lab, your towers break through the 84.00 mm ceiling into mixed numbers.

STEP-BY-STEP APPARATUS SETUP:

1. Stand the Whole Frame [1/1] (84.00 mm) on your desk.
2. Stack five [1/4] blocks in a single tower.
3. Notice: Your tower extends above the 84.00 mm frame!

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks five [1/4]s, two [1/1] wholes, and [1/2] blocks

DATUM TARGET: 5/4 = 5 × 21.00 = 105.00 mm = 84.00 mm (1 Whole) + 21.00 mm (1/4)

CALCULATOR KEY: [5] [/] [4] [a b/c] -> 1 1/4

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 4A, Chapter 5 (Improper Fractions)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 4, Unit 4 (Fractions Greater Than 1)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 5, Ch. 2 • CCSS.MATH.4.NF.B.3.B • TEKS 4.3B

GRADE LEVEL BAND: Grades 4–6 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 120–126

HARDWARE UNBOXING BOOKLET: FRAC-01 Booklet: Page 4 (Fill Datum) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: BREAKING THE CEILING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Measure how much your five [1/4] tower extends above the 84.00 mm whole frame: exactly 21.00 mm.

Step 2: Stand a loose [1/4] block on top of the whole frame. Notice: It levels with your tower!

Step 3: Record: 5/4 is the exact same physical amount as 1 Whole + 1/4 (1 1/4).

Step 4: Now stack seven [1/3] blocks. How many whole frames does it fill? (2 Wholes + 1/3 = 2 1/3 = 7/3).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
5/4 = 5 × 21.00 = 105.00 mm = 84.00 mm (1 Whole) + 21.00 mm (1/4)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks five [1/4]s, two [1/1] wholes, and [1/2] blocks

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: DECOMPOSING IMPROPER FRACTIONS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you convert 11/8 into a mixed number using only blocks, and verify it on your calculator?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Stack eleven [1/8] blocks. Compare against the 84.00 mm whole.**
   Expected Physical Finding: **Tower height = 115.50 mm**

2. **How many whole frames are filled, and what is the remainder?**
   Expected Physical Finding: **1 Whole (8/8) + 3/8 remainder**

3. **Key [11] [ENTER] [8] [/] on SC-6.**
   Expected Physical Finding: **1.375 = 1 3/8**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [5] [/] [4] [a b/c] -> 1 1/4

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
5	X: 5	Numerator 5
4	Y: 5 | X: 4	Denominator 4
/	X: 1.25	Decimal evaluation
a b/c	X: 1_1/4	Convert to mixed number 1 and 1/4

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Rational Stacks & RPN Operations

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did dividing 3/4 by 1/8 give a whole number answer of 6?'
- Ask: 'How is division with fractions just asking: how many times does block B fit in block A?'
- Ask: 'How does the RPN stack keep track of multi-step fraction problems?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to invert and multiply without understanding why. Ground them in the physical question: 'How m
any small blocks fit in the big block?'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Solve (1/2 + 1/3) ÷ 5/12 on the RPN stack. What is the final answer?"
""

OFFICIAL SOLUTION KEY:

1/2 + 1/3 = 5/6 = 10/12. (10/12) ÷ (5/12) = 2.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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step-by-step video solution, and printable rubric.

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FACILITATOR / TEACHER SIDE • FRAC-01 Booklet: Page 7 (RPN Stack Mapping) & Page 8 (Exercises)

CALIFORNIA DESIGN LAB • VERIFIED REPRODUCIBLE

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Rational Stacks & RPN Operations

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

When you solve fraction arithmetic on paper, you must remember flip-and-multiply rules. In the RPN calculator, operations execute exactly like stacking and unsta

STEP-BY-STEP APPARATUS SETUP:

1. Place Block [3/4] on your desk. On top of it, place Block [2/3].
2. Key [3] [.] [4] into your SC-6 or SC-32 calculator. Press [ENTER].
3. Key [2] [.] [3] into Register X. Press [*].

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Full Fraction Kit + SC-6 Elementary / SC-32 Scientific Calculator

DATUM TARGET: Direct Isomorphism: Top of Physical Tower = Register X, Bottom = Register Y

CALCULATOR KEY: [3] [.] [4] [ENTER] [2] [.] [3] [*] -> 1/2

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 6A, Chapter 3 (Fractions & Decimals Operations)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 5 (Multiplying & Dividing Fractions)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 6, Ch. 2 • CCSS.MATH.6.NS.A.1 • TEKS 6.3B
GRADE LEVEL BAND:	Grades 5–8 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 126–134
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 7 (RPN Stack Mapping) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
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CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

STUDENT ENTRANCE SIDE • Fraction Studio Proportional Towers • Grades 5–8

CALIFORNIA LAB QC SPECIFICATION // STACKCALC

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: FRACTION MULTIPLICATION AS SCALING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at the result on your calculator LCD: 1/2 (or 0.50).

Step 2: Stand Block [1/2] beside your desk. Why is 2/3 of 3/4 equal to 1/2?

Step 3: Three [1/4] blocks divided into three groups gives 1/4 per group. Two of those groups equal two [1/4]s = 1/2!

Step 4: The physical blocks confirm the RPN stack calculation: scaling 3/4 by 2/3 shrinks it to exactly 1/2.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Direct Isomorphism: Top of Physical Tower = Register X, Bottom = Register Y

PHYSICAL APPARATUS MANIFEST:

Full Fraction Kit + SC-6 Elementary / SC-32 Scientific Calculator

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: FRACTION DIVISION WITHOUT FLIPPING

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 3/4 ÷ 1/8 using blocks. How many [1/8] blocks fit inside Block [3/4]?

FORM A MASTER STATION TASKS (Complete All 3):

1 **Stack [1/8] blocks beside Block [3/4] until they level.**
Expected Physical Finding: **Six [1/8] blocks level to 3/4**

2 **Execute on SC-6: [3] [.] [4] [ENTER] [1] [.] [8] [/].**
Expected Physical Finding: **Result = 6 (Exact integer!)**

3 **Explain why 3/4 ÷ 1/8 = 6 without using paper rules.**
Expected Physical Finding: **Six eighths fit inside three fourths.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [.] [4] [ENTER] [2] [.] [3] [*] -> 1/2

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3.4	X: 3/4	Key fraction 3/4
ENTER	Y: 3/4 | X: 3/4	Stack lift: 3/4 moves to Y
2.3	Y: 3/4 | X: 2/3	Key fraction 2/3 into X
*	X: 1/2	Multiply: (3/4) × (2/3) = 6/12 = 1/2

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Multiplication Mountain

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does adding physical millimeters multiply the numbers together?'
- Ask: 'Could you build 14 using only two blocks? Which blocks must you choose?'
- Ask: 'What happens to the tower if you try to make 17 out of smaller blocks?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to add the numerals (2 + 3 = 5) instead of reading the product. Direct their attention to the level comparison with Block [6], not Block [5].

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a mystery tower made of three prime blocks. My summit levels exactly with Composite Block [42]. What three pri"
"me blocks am I?"

OFFICIAL SOLUTION KEY:

Blocks [2], [3], and [7] (since 2 × 3 × 7 = 42).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Multiplication Mountain

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Can you build multiplication with physical height? In this lab, you discover why multiplying numbers is identical to adding physical tower heights.

STEP-BY-STEP APPARATUS SETUP:

1. Pull Prime Blocks [2] (Teal) and [3] (Coral) from your Gridfinity storage tray.
2. Snap Block [3] on top of Block [2] to create a two-block tower.
3. Pull Composite Block [6] (Amber) and place it side-by-side with your tower on the desk.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Prime Blocks [2], [3], [5] and Composite Blocks [6], [10], [15] + Stage

DATUM TARGET: H(2)=7.00mm + H(3)=11.09mm = H(6)=18.09mm (K_log = 23.2535 mm/dec)

CALCULATOR KEY: [2] [ENTER] [3] [×] -> 6

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 4A, Chapter 3 (Factors & Multiples)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 4, Unit 2 (Factors & Prime Numbers)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 6, Sec. 1.2 • CCSS.MATH.4.OA.B.4 • TEKS 5.4A

GRADE LEVEL BAND: Grades 4–6 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 7 ('Primes & Factor Towers'), pp. 142–156

HARDWARE UNBOXING BOOKLET: PRIME-01 Booklet: Page 2 (Log Invariant) & Page 5 (Tutorial 1)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE LEVEL SUMMIT TEST

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Lay a straightedge across the summits of Tower [2]+[3] and Block [6]. Notice they level perfectly.

Step 2: Pull Prime Block [5] and snap it on top of Block [2]. Find Composite Block [10] to verify the match.

Step 3: Measure Block [3] and Block [5] stacked. Verify it levels horizontally with Block [15].

Step 4: Record your discovery: Every time you stack prime blocks, their heights ADD together to form the product.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
H(2)=7.00mm + H(3)=11.09mm = H(6)=18.09mm (K_log = 23.2535 mm/dec)

PHYSICAL APPARATUS MANIFEST:

Prime Blocks [2], [3], [5] and Composite Blocks [6], [10], [15] + Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE THREE-BLOCK PEAK

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you build the number 30 using only prime blocks? How many different orders can you stack them in, and does the final height ever change?

FORM A MASTER STATION TASKS (Complete All 3):

1 **Stack [2] + [3] + [5]. Measure total height against Block [30].**
Expected Physical Finding: Sum = 34.41 mm (Exact Match)

2 **Stack [5] + [2] + [3] (different order). Does the summit change?**
Expected Physical Finding: Identical height: 34.41 mm

3 **Execute on SC-6: [2] [ENTER] [3] [×] [5] [×]. What is Register X?**
Expected Physical Finding: 30

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [ENTER] [3] [×] -> 6

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
2	X: 2	Key first prime factor into register X
ENTER	Y: 2 | X: 2	Stack lift copies 2 into register Y
3	Y: 2 | X: 3	Key second prime factor into register X
×	X: 6	Multiply: registers collapse into product 6

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: The Prime Fingerprint

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Is it possible for any composite number to have two different sets of prime blocks?'
- Ask: 'Why are prime numbers called the bedrock of mathematics?'
- Ask: 'Can you build a tower for 25 using only [2] and [3] blocks? Why not?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students treating 9 as a prime block. Remind them to check if Block [9] can be replaced by two [3] blocks.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A mystery number has the prime fingerprint [2] × [2] × [2] × [3] × [5]. What is the number, and what is its total t"
"ower height?"

OFFICIAL SOLUTION KEY:

Number = 120. Tower Height = 3(7.00) + 11.09 + 16.25 = 48.34 mm.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: The Prime Fingerprint

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Every integer greater than 1 has a unique secret code made exclusively of prime numbers. No matter what path you take, you always end up with the exact same pri

STEP-BY-STEP APPARATUS SETUP:

1. Pull Composite Block [36] from your kit and place it in the center of the table.
2. Team Member A builds 36 using factor pair 4 × 9 (Stack Block [4] and Block [9]).
3. Team Member B builds 36 using factor pair 6 × 6 (Stack Block [6] and Block [6]).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Prime Blocks [2], [2], [3], [3], [5] and Composite Blocks [12], [18], [36]
DATUM TARGET:	H(36) = 2·H(2) + 2·H(3) = 14.00 + 22.18 = 36.18 mm
CALCULATOR KEY:	[2] [x²] [3] [x²] [×] -> 36

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 5A, Chapter 1, Lesson 1.2 (Prime Factorization)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 1, Module 3 (Factor Trees)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 6, Sec. 1.3 • CCSS.MATH.6.NS.B.4 • TEKS 6.7A
GRADE LEVEL BAND:	Grades 5–7 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7 ('The Prime Fingerprint'), pp. 146–152
HARDWARE UNBOXING BOOKLET:	PRIME-01 Booklet: Page 2 (Uniqueness Theorem) & Page 6 (Aha! Moment)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: DECOMPOSING TO BEDROCK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Compare Tower 4 × 9 and Tower 6 × 6. Confirm both level perfectly to Block [36].

Step 2: Decompose Block [4] into two [2] blocks, and Block [9] into two [3] blocks.

Step 3: Decompose both [6] blocks from Team B's tower into [2] and [3].

Step 4: Count bedrock blocks in both piles: Both hold exactly two [2]s and two [3]s ($2^2 \times 3^2$).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
$H(36) = 2 \cdot H(2) + 2 \cdot H(3) = 14.00 + 22.18 = 36.18$ mm

PHYSICAL APPARATUS MANIFEST:

Prime Blocks [2], [2], [3], [3], [5] and Composite Blocks [12], [18], [36]

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: FACTOR TREE DECONSTRUCTION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you build Composite Block [60] using three different factor pairs, and prove all three reduce to the exact same prime blocks?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Path 1: Start with 6 × 10. Decompose into primes.**
   Expected Physical Finding: [2] + [3] + [2] + [5]

2. **Path 2: Start with 4 × 15. Decompose into primes.**
   Expected Physical Finding: [2] + [2] + [3] + [5]

3. **Path 3: Start with 2 × 30. Decompose into primes.**
   Expected Physical Finding: [2] + [2] + [3] + [5]

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [x^2] [3] [x^2] [×] -> 36

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
2	X: 2	Key base prime 2
x^2	X: 4	Square prime: $2^2 = 4$ (Tower of two [2] blocks)
3	Y: 4 | X: 3	Key base prime 3
x^2	Y: 4 | X: 9	Square prime: $3^2 = 9$ (Tower of two [3] blocks)
×	X: 36	Multiply: 4 × 9 = 36

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: The Greatest Common Factor

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'If two numbers share no prime blocks in common, what is their GCF?'
- Ask: 'Why can't the GCF ever be taller than the shortest original tower?'
- Ask: 'How does the comparator eliminate guess-and-check errors?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students including all blocks instead of only shared pairs. Emphasize that GCF requires an exact 1-to-1 matchi
ng exchange between both towers.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"Two numbers have a GCF of 14. One number is 42. What is the smallest possible value for the other number?"
""

OFFICIAL SOLUTION KEY:

Smallest other number is 14 (or 28, 56, etc.).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: The Greatest Common Factor

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Finding the Greatest Common Factor on paper takes long lists of factors. On the bilateral comparator stage, you simply slide the towers together and pick out the

STEP-BY-STEP APPARATUS SETUP:

1. Dock the bilateral slide rule stage onto the center of your table.
2. Build Tower A for 24 using prime blocks: [2], [2], [2], [3].
3. Build Tower B for 36 using prime blocks: [2], [2], [3], [3].

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Prime Blocks [2], [2], [2], [3], [3] + Bilateral Slide Rule Comparator Stage
DATUM TARGET:	GCF(24, 36) = $2^2 \times 3 = 12$ -> Height = 2(7.00) + 11.09 = 25.09 mm
CALCULATOR KEY:	[24] [ENTER] [36] [GCF] -> 12

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 5A, Chapter 1, Lesson 1.3 (Greatest Common Factor)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 1, Module 3, Session 2 (GCF)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 6, Sec. 1.4 • CCSS.MATH.6.NS.B.4 • TEKS 6.7A
GRADE LEVEL BAND:	Grades 5–7 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 152–158
HARDWARE UNBOXING BOOKLET:	PRIME-01 Booklet: Page 5 (Comparator Setup) & Page 6 (GCF Proof)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE OVERLAP EXTRACTION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Place Tower A in the left comparator well and Tower B in the right well.

Step 2: Match common blocks pair-by-pair: Pull one [2] from both. Pull a second [2] from both. Pull one [3] from both.

Step 3: What remains? Tower A has an extra [2]; Tower B has an extra [3]. These do not match.

Step 4: Stack the shared matching blocks ([2] + [2] + [3]). Measure this tower against Block [12]. You found the GCF!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
GCF(24, 36) = $2^2 \times 3 = 12$ –> Height = 2(7.00) + 11.09 = 25.09 mm

PHYSICAL APPARATUS MANIFEST:

Prime Blocks [2], [2], [2], [3], [3] + Bilateral Slide Rule
Comparator Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THREE-NUMBER COMPARISON

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you use prime blocks to find the GCF of 30, 45, and 75 without writing a single calculation on paper?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Prime factorize 30, 45, and 75 into block piles.**
   Expected Physical Finding: 30=[2,3,5], 45=[3,3,5], 75=[3,5,5]

2. **Identify the blocks common to ALL THREE piles.**
   Expected Physical Finding: One [3] and one [5]

3. **What is GCF(30, 45, 75)? Verify with Block [15].**
   Expected Physical Finding: 15 (Height = 27.34 mm)

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [24] [ENTER] [36] [GCF] –> 12

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
24	X: 24	Key first number into register X
ENTER	Y: 24 | X: 24	Stack lift
36	Y: 24 | X: 36	Key second number into register X
GCF	X: 12	Extract greatest common divisor from stack

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: The Least Common Multiple

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is the LCM always as large as or larger than both starting numbers?'
- Ask: 'If two runners lap a track every 6 and 8 minutes, when do they cross the start line together?'
- Ask: 'How does taking the prime union guarantee the smallest leveling summit?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking LCM means multiplying 6 × 8 = 48. Show them that 24 levels first, making 24 the LEAST common multiple.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Traffic light A flashes every 12 seconds. Traffic light B flashes every 18 seconds. Traffic light C flashes every 3"
"0 seconds. When do all three flash together?"

OFFICIAL SOLUTION KEY:

LCM(12, 18, 30) = $2^2 \times 3^2 \times 5 = 180$ seconds (3 minutes).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: The Least Common Multiple

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

When will two repeating rhythmic patterns clash together at the exact same beat? In this lab, you build stepping towers of 6 and 8 until their summits level off.

STEP-BY-STEP APPARATUS SETUP:

1. Pull several Block [6]s and Block [8]s from your storage tray.
2. Team Member A stacks blocks of 6: 6, 12, 18, 24, 30...
3. Team Member B stacks blocks of 8: 8, 16, 24, 32...

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Prime Blocks [2], [2], [2], [3], [3] + Multiples of 6 and 8
DATUM TARGET:	LCM(6, 8) = $2^3 \times 3 = 24$ -> Height = 3(7.00) + 11.09 = 32.09 mm
CALCULATOR KEY:	[6] [ENTER] [8] [LCM] -> 24

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 6A, Chapter 1, Lesson 1.4 (Least Common Multiple)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 1, Module 3, Session 3 (LCM)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 6, Sec. 1.5 • CCSS.MATH.6.NS.B.4 • TEKS 6.7A
GRADE LEVEL BAND:	Grades 5–7 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 156–162
HARDWARE UNBOXING BOOKLET:	PRIME-01 Booklet: Page 2 (LCM Formula) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE SUMMIT LEVELING RACE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Build Tower 6 (height 18.09mm) and Tower 8 (height 21.00mm). Notice Tower 8 is taller.

Step 2: Add another [6] to Tower 6. Now Tower 6 is 12 (height 25.09mm) and taller than 8.

Step 3: Add another [8] to Tower 8. Tower 8 is 16 (height 28.00mm). Keep going!

Step 4: Stop when both summits reach the exact same horizontal level: 24 (height 32.09mm). You found LCM(6,8)!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
LCM(6, 8) = $2^3 \times 3 = 24$ –> Height = 3(7.00) + 11.09 = 32.09 mm

PHYSICAL APPARATUS MANIFEST:

Prime Blocks [2], [2], [2], [3], [3] + Multiples of 6 and 8

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE PRIME UNION SHORTCUT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Instead of stacking composite blocks, can you find LCM(12, 18) in 10 seconds by taking the UNION of their prime blocks?

FORM A MASTER STATION TASKS (Complete All 3):

1 **Factor 12: [2, 2, 3]. Factor 18: [2, 3, 3].**
Expected Physical Finding: **Blocks: two [2]s, two [3]s**

2 **Combine the maximum count of each prime: [2, 2, 3, 3].**
Expected Physical Finding: **$2^2 \times 3^2 = 36$**

3 **Verify: Does 36 level both Tower 12 and Tower 18?**
Expected Physical Finding: **Yes! 12×3=36 and 18×2=36.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [6] [ENTER] [8] [LCM] –> 24

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
6	X: 6	Key first number
ENTER	Y: 6 | X: 6	Stack lift
8	Y: 6 | X: 8	Key second number
LCM	X: 24	Calculate least common multiple

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: The Product-Balance Theorem

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Where did the blocks from Tower A and Tower B go when we made GCF and LCM?'
- Ask: 'Can any prime block be created or destroyed during this regrouping?'
- Ask: 'Why does this prove that GCF(A,B) × LCM(A,B) = A × B for ALL numbers?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking this is an approximation. Emphasize that because prime blocks are physical conserved units, this is an exact geometric proof.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Two numbers have a product of 1,200. Their GCF is 10. Without knowing what the numbers are, what is their LCM?"
""

OFFICIAL SOLUTION KEY:

LCM = 1,200 / 10 = 120.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: The Product-Balance Theorem

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Is it a coincidence that multiplying the GCF and LCM always equals the product of the original two numbers? In this lab, you prove this universal theorem using p

STEP-BY-STEP APPARATUS SETUP:

1. Choose numbers A = 12 and B = 18.
2. Build Tower A (12 = [2]+[2]+[3]) and Tower B (18 = [2]+[3]+[3]).
3. Stack Tower A on top of Tower B to create the 'Combined Product Tower' (12 × 18).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Prime Blocks [2], [2], [2], [3], [3] for numbers 12 and 18 + Comparator Stage
DATUM TARGET:	H(12) + H(18) = H(GCD) + H(LCM) = H(6) + H(36) = 54.27 mm
CALCULATOR KEY:	[12] [ENTER] [18] [×] [12] [ENTER] [18] [GCF] [12] [ENTER] [18] [LCM] [×] [−] -> 0

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 6A, Chapter 1, Lesson 1.5 (Advanced Number Theory)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 1, Module 3 (Summit Balances)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 6, Ch. 1 Review • CCSS.MATH.6.NS.B.4 • TEKS 6.7A
GRADE LEVEL BAND:	Grades 6–8 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 7 ('Brent Solver & Number Theory'), pp. 88–96
HARDWARE UNBOXING BOOKLET:	PRIME-01 Booklet: Page 2 (Theorem 1.4) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE PRIME CONSERVATION PROOF

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Pull out the GCF blocks shared by both numbers: one [2] and one [3] (Tower GCF = 6).

Step 2: Collect all the remaining blocks into the LCM pile: [2], [2], [3], [3] (Tower LCM = 36).

Step 3: Stack Tower GCF (6) on top of Tower LCM (36) to create the 'GCF × LCM Tower'.

Step 4: Stand the Combined Product Tower beside the GCF × LCM Tower: Both towers contain the exact same five prime blocks! They level perfectly!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
H(12) + H(18) = H(GCD) + H(LCM) = H(6) + H(36) = 54.27 mm

PHYSICAL APPARATUS MANIFEST:

Prime Blocks [2], [2], [2], [3], [3] for numbers 12 and 18 +
Comparator Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE ALGEBRAIC EQUALITY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Test this invariant with numbers 20 and 30. Does H(20) + H(30) equal H(GCF) + H(LCM)?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Find GCF(20, 30) and LCM(20, 30).**
   Expected Physical Finding: GCF = 10, LCM = 60

2. **Calculate 20 × 30 and 10 × 60.**
   Expected Physical Finding: Both equal 600

3. **Count prime blocks: 20=[2,2,5], 30=[2,3,5]. What are GCF and LCM primes?**
   Expected Physical Finding: GCF=[2,5], LCM=[2,2,3,5]. Total primes conserved!

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [12] [ENTER] [18] [×] [12] [ENTER] [18] [GCF] [12] [ENTER] [18] [LCM] [×] [−] -> 0

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
12	X: 12	Key A = 12
ENTER	Y: 12 | X: 12	Stack lift
18	Y: 12 | X: 18	Key B = 18
×	X: 216	A × B = 216
6	Y: 216 | X: 6	Key GCF(12, 18) = 6
36	Y: 216 | X: 6 | X: 36	Key LCM(12, 18) = 36

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Exponents, Stacks & Cryptography

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did finding 13 × 17 take only 1 second, but finding the factors of 221 took multiple minutes?'
- Ask: 'What happens when the prime numbers have 300 digits instead of 2 digits?'
- Ask: 'How does the RPN stack mirror building a tower of blocks?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students testing even numbers like 8 or 10. Remind them: if a number is not divisible by 2, it CANNOT be divis
ible by any multiple of 2!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a semiprime number (product of exactly two primes). My smaller prime factor is 11. My total tower height is 38"
".64 mm. What number am I?"

OFFICIAL SOLUTION KEY:

H(11) = 24.21 mm. Remaining height = 38.64 − 24.21 = 14.43 mm -> H(13) = 25.90 mm... Testing: 11 × 23 = 253.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Exponents, Stacks & Cryptography

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Modern internet banking and cybersecurity (RSA cryptography) are built on a simple secret: multiplying two prime numbers together is instant, but factoring them

STEP-BY-STEP APPARATUS SETUP:

1. Pull Prime Blocks [17] and [19] from your storage tray.
2. Multiply them together on your SC-32 calculator: [17] [ENTER] [19] [×]. (Result: 323).
3. Challenge: If someone gives you the number 323, how hard is it to find the two prime blocks without a computer?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Full Prime Kit (Blocks 2, 3, 5, 7, 11, 13, 17, 19) + SC-32 Hardware
DATUM TARGET:	Prime Tower H(p^k) = k · H(p) (Direct physical multiplication of height)
CALCULATOR KEY:	[2] [ENTER] [5] [y^x] -> 32

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7A, Chapter 1 (Prime Numbers & Exponents)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 8 (Culminating Math Projects)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Accelerated Grade 7, Ch. 1 • CCSS.MATH.8.EE.A.1 • TEKS 8.2D
GRADE LEVEL BAND:	Grades 6–8+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 10 ('Computer Science, Bases & Ciphers'), pp. 160–174
HARDWARE UNBOXING BOOKLET:	PRIME-01 Booklet: Page 7 (RPN Stack Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: EXPONENTIAL REPETITION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Build a tower of five [2] blocks (2^5). Measure height: 5 × 7.00 = 35.00 mm.

Step 2: Key [2] [ENTER] [5] [y^x] into SC-32. Verify Register X reads 32.

Step 3: Build a tower of three [3] blocks ($3^3 = 27$). Height = 3 × 11.09 = 33.27 mm.

Step 4: Combine $2^3 \times 3^2$ (three [2]s and two [3]s). Verify Register X reads 72.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

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✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Prime Tower H(p^k) = k · H(p) (Direct physical multiplication of height)

PHYSICAL APPARATUS MANIFEST:

Full Prime Kit (Blocks 2, 3, 5, 7, 11, 13, 17, 19) + SC-32 H
ardware

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE RSA ONE-WAY FUNCTION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you find the two prime factor blocks that multiply to make 221? (Hint: Neither factor is 2, 3, or 5).

FORM A MASTER STATION TASKS (Complete All 3):

1. **Test prime 7: Does 221 divide by 7?**
   Expected Physical Finding: **221 ÷ 7 = 31.57 (No)**

2. **Test prime 11: Does 221 divide by 11?**
   Expected Physical Finding: **221 ÷ 11 = 20.09 (No)**

3. **Test prime 13: Does 221 divide by 13?**
   Expected Physical Finding: **221 ÷ 13 = 17 (Exact match: 13 × 17 = 221!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [ENTER] [5] [y^x] -> 32

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
2	X: 2	Base prime 2
ENTER	Y: 2 | X: 2	Stack lift
5	Y: 2 | X: 5	Exponent 5 (Stack of five [2] blocks)
y^x	X: 32	Evaluate $2^5 = 32$

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Uniform 10-Sector Trial

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why doesn\'t each number land exactly 5 times in 50 spins?'
- Ask: 'What makes the 36° sectors geometrically fair?'
- Ask: 'How does the captive retention barb keep the arrow from wobbling off course?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking a number is 'due' to hit if it hasn't appeared yet (Gambler's Fallacy). Remind them: the spi
nner has no memory!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"What is the theoretical probability of spinning a prime number (2, 3, 5, 7) on the 1–10 dial?"
""

OFFICIAL SOLUTION KEY:

Four primes out of 10 numbers: P(Prime) = 4/10 = 2/5 = 40%.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/dice-and-spinner/#product-videos

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Uniform 10-Sector Trial

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Is a spinner truly random? In this lab, test the low-friction captive spindle over 50 spins and compare your experimental results to the mathematical truth.

STEP-BY-STEP APPARATUS SETUP:

1. Snap the 10-Sector Uniform Dial onto the spinner base.
2. Give the arrow a test flick: Notice it spins freely on its conical bearing for >5 seconds.
3. Prepare a tally chart for numbers 1 through 10.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Modular Spinner Base + 10-Sector Uniform Dial (36° each, 1–10) + Captive Spindle
DATUM TARGET:	Fairness Invariant: Each Sector Sweeps Exactly 36.0° (P(n) = 1/10 = 0.1000 = 10%)
CALCULATOR KEY:	[1] [/] [10] -> 0.10 (10% per sector)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 6B, Chapter 13 (Theoretical Probability)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7 (Probability & Fair Games)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.1 • CCSS.MATH.7.SP.C.5 • TEKS 7.6I
GRADE LEVEL BAND:	Grades 5–7 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7 ('Data & Probability'), pp. 158–166
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 2 (Fairness Geometry) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 50-SPIN EXPERIMENTAL RUN

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Spin the arrow 50 times. Record each landing number in your tally chart.

Step 2: Calculate theoretical probability: 1 out of 10 = 10% (expected 5 landings per number in 50 spins).

Step 3: Compare your actual results: Did any number land exactly 5 times? Did any land 7 or 8 times?

Step 4: Conclude: In a small sample (50 spins), experimental results fluctuate around the theoretical average.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Fairness Invariant: Each Sector Sweeps Exactly 36.0° (P(n) = 1/10 = 0.1000 = 10%)

PHYSICAL APPARATUS MANIFEST:

Modular Spinner Base + 10-Sector Uniform Dial (36° each, 1–10) + Captive Spindle

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: CALCULATING RELATIVE FREQUENCY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Calculate the relative frequency of spinning an even number (2, 4, 6, 8, 10).

FORM A MASTER STATION TASKS (Complete All 3):

1 **What is the theoretical probability of spinning an EVEN number?**
Expected Physical Finding: **5/10 = 1/2 = 50%**

2 **Count your team's total even landings out of 50 spins.**
Expected Physical Finding: **Typically 22 to 28 landings (44% to 56%)**

3 **What happens if you combine data with all other tables in the room (300 spins)?**
Expected Physical Finding: **Relative frequency moves much closer to exactly 50%!**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[1] [/] [10] -> 0.10 (10% per sector)**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Target outcome count
10	Y: 1 | X: 10	Total equal sector sample space
/	X: 0.10	Theoretical probability evaluation

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: Weighted Realities: The Biased Dial

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does Zone A cover half the circle?'
- Ask: 'How do casinos design games where you win small prizes often but lose money over time?'
- Ask: 'What would the angle of Zone D need to be to give a 1-in-3 chance?' (120°).

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 4 sectors means each has a 1/4 chance. Point directly to the physical sizes of the wedges!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If a new sector E has an angle of 72°, what percentage of the board does it occupy?"
""

OFFICIAL SOLUTION KEY:

72° / 360° = 1/5 = 20%.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/dice-and-spinner/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: Weighted Realities: The Biased Dial

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Not all events in life have equal odds. Weather forecasts, carnival games, and insurance rates use weighted probabilities. In this lab, test an unfair spinner.

STEP-BY-STEP APPARATUS SETUP:

1. Flip the dial to Side B (the Non-Uniform Weighted Dial).
2. Notice the four colored zones: Zone A (Blue, 180°), Zone B (Green, 90°), Zone C (Amber, 54°), Zone D (Red, 36°).
3. Prepare to record 40 spins.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Modular Spinner Base + Side B Weighted Dial (Sector A: 180°, B: 90°, C: 54°, D: 36°)

DATUM TARGET: Angular Invariant: Sector Angle / 360° = Theoretical Probability (180° = 50%, 90° = 25%)

CALCULATOR KEY: [180] [/] [360] -> 0.50 (50% Zone)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

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SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7B, Chapter 9 (Experimental Probability)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7, Module 2 (Unfair Spinners)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.2 • CCSS.MATH.7.SP.C.7.B • TEKS 7.6I
GRADE LEVEL BAND:	Grades 6–8 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 166–172
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 5 (Side B Weighted Dial) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

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2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: THE CARNIVAL SIMULATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Calculate theoretical odds: A = 180/360 = 50%; B = 90/360 = 25%; C = 54/360 = 15%; D = 36/360 = 10%.

Step 2: Spin the arrow 40 times. Tally how many times it lands in Zone A, B, C, and D.

Step 3: Compare: Did Zone A land roughly twice as often as Zone B? Did Zone D land least often?

Step 4: Conclude: The probability of landing in any sector is DIRECTLY proportional to its central angle.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Angular Invariant: Sector Angle / 360° = Theoretical Probability (180° = 50%, 90° = 25%)

PHYSICAL APPARATUS MANIFEST:

Modular Spinner Base + Side B Weighted Dial (Sector A: 180°,
B: 90°, C: 54°, D: 36°)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: DESIGNING A CARNIVAL GAME

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

If it costs $1 to spin: Landing on D wins $5. Landing on C wins $2. Landing on A or B wins nothing.

FORM A MASTER STATION TASKS (Complete All 3):

1 **Calculate expected payout: P(D)×$5 + P(C)×$2.**
Expected Physical Finding: **(0.10 × $5) + (0.15 × $2) = $0.50 + $0.30 = $0.80**

2 **If players pay $1.00 and receive $0.80 on average, who wins long term?**
Expected Physical Finding: **The carnival owner makes $0.20 per spin!**

3 **Is this a fair game?**
Expected Physical Finding: **No, it is mathematically biased toward the house.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[180] [/] [360] -> 0.50 (50% Zone)**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
180	X: 180	Sector A arc degrees
360	Y: 180 | X: 360	Full circle degrees
/	X: 0.50	Sector probability 50%

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: The Law of Large Numbers

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why are small surveys unreliable in medicine or politics?'
- Ask: 'How does equal-volume pip displacement keep the die from landing on 6 too often?'
- Ask: 'What would happen if we rolled 1,000,000 times?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking the die 'tries' to balance itself out. Emphasize that large numbers work through averaging, not through magical compensation.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A pollster surveys 10 voters and 7 support Candidate A. Why would a statistician say this sample is too small?"
""

OFFICIAL SOLUTION KEY:

A small sample has enormous variance (margin of error ±30%). At least 500–1,000 voters are needed for accuracy.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

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PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: The Law of Large Numbers

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does flipping a coin 4 times sometimes give 4 heads, but flipping it 1,000 times almost never gives 1,000 heads? In this lab, observe the Law of Large Numbe

STEP-BY-STEP APPARATUS SETUP:

1. Take one 18mm precision die from your kit.
2. Notice the equal-volume debossed pips: Face 6 uses smaller holes than Face 1 so all faces have equal weight!
3. Prepare to track the appearance of rolling a '6'.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Four 18mm Precision Fair Dice + Dice Landing Tray

DATUM TARGET: Convergence Invariant: Lim(n->∞) |f_n − P| = 0 (Relative frequency converges to theoretical truth)

CALCULATOR KEY: [1] [/] [6] -> 0.1667 (16.67% per face)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
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BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7, Module 3 (Sample Size & Predictions)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.3 • CCSS.MATH.7.SP.C.6 • TEKS 7.6D
GRADE LEVEL BAND:	Grades 6–8 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 9 ('Statistics & Curve Fitting'), pp. 142–156
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 6 (Aha! Discovery: Law of Large Numbers)

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3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: SAMPLE SIZE TRACKING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Trial 1 (Small Sample): Roll the die 6 times. Did 6 appear exactly once? (Often 0 or 2 times).

Step 2: Trial 2 (Medium Sample): Roll 30 times. Count total 6s. Calculate percentage (expected ~16.7%).

Step 3: Trial 3 (Large Class Aggregate): Pool data from all 6 tables in the classroom (total 180 rolls).

Step 4: Notice: As the number of rolls grows from 6 -> 30 -> 180, the percentage of 6s locks onto 16.7%!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Convergence Invariant: Lim(n->∞) |f_n - P| = 0 (Relative frequency converges to theoretical truth)

PHYSICAL APPARATUS MANIFEST:

Four 18mm Precision Fair Dice + Dice Landing Tray

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE ERROR MARGIN SQUEEZE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Look at how the error between experimental and theoretical shrinks.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Theoretical rate for rolling a 6.**
   Expected Physical Finding: **1/6 = 16.67%**

2. **In 6 rolls, getting two 6s gives what percentage?**
   Expected Physical Finding: **2/6 = 33.3% (Large error of +16.7%)**

3. **In 600 rolls, getting 105 6s gives what percentage?**
   Expected Physical Finding: **105/600 = 17.5% (Small error of only +0.8%)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [/] [6] -> 0.1667 (16.67% per face)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Single face
6	Y: 1 | X: 6	Total fair die faces
/	X: 0.1667	Theoretical probability = 16.67%

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 4: The 2d6 Triangular Sum Distribution

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is sum 7 six times more likely than sum 2?'
- Ask: 'Why do board game designers put robber/penalty tiles on 7?'
- Ask: 'How does rolling three dice (3d6) turn this triangle into a smooth bell curve?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 2 and 7 have equal chances because they are both numbers on the chart. Remind them: 7 has 6 combinations, but 2 has only 1!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"What is the probability of rolling a sum GREATER than 9 on two dice?"
""

OFFICIAL SOLUTION KEY:

Sums 10 (3 ways), 11 (2 ways), 12 (1 way). Total = 3 + 2 + 1 = 6 ways -> 6/36 = 1/6.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 1: The 2d6 Triangular Sum Distribution

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is 7 the most famous number in board games like Settlers of Catan or Craps? In this lab, discover why adding two fair dice creates a triangular bell curve.

STEP-BY-STEP APPARATUS SETUP:

1. Take two different colored dice: Die A (Teal) and Die B (Amber).
2. List all possible sums: Minimum sum is 1+1 = 2; Maximum sum is 6+6 = 12.
3. Prepare a dot plot chart for sums 2 through 12.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Two 18mm Fair Dice (Teal & Amber) + Sum Tracking Sheet

DATUM TARGET: Combinatoric Peak: 36 Outward Outcomes -> Sum 7 Has 6 Ways (P=6/36=1/6), Sum 2 Has Only 1 Way (P=1/36)

CALCULATOR KEY: [6] [/] [36] -> 0.1667 (16.67% for Sum 7)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

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SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 12, Lesson 12.4 (Compound Events)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7, Module 4 (Dice Sums & Bell Curves)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 10, Sec. 10.4 • CCSS.MATH.7.SP.C.8 • TEKS 7.6I
GRADE LEVEL BAND:	Grades 6–8+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 170–178
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 2 (2d6 Distribution Table) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 2: LAB INVESTIGATION: THE 36-COMBINATION MAP

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: How many ways can you roll a sum of 2? Only ONE way: (1, 1). P(2) = 1/36.

Step 2: How many ways to roll a sum of 3? TWO ways: (1, 2) and (2, 1). P(3) = 2/36.

Step 3: Count ways to roll 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). Exactly SIX ways! P(7) = 6/36 = 1/6.

Step 4: Roll the two dice 36 times and plot the sums. Look at the shape: It forms a perfect triangular peak centered at 7!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Combinatoric Peak: 36 Outward Outcomes -> Sum 7 Has 6 Ways (P=6/36=1/6), Sum 2 Has Only 1 Way (P=1/36)

PHYSICAL APPARATUS MANIFEST:

Two 18mm Fair Dice (Teal & Amber) + Sum Tracking Sheet

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 6–8+

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE PROBABILITY SYMMETRY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Observe the symmetric drop on both sides of 7.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Compare P(6) and P(8): How many ways to make each?**
   Expected Physical Finding: **Both have exactly 5 ways (5/36 each)**

2. **Compare P(5) and P(9): How many ways to make each?**
   Expected Physical Finding: **Both have exactly 4 ways (4/36 each)**

3. **Compare P(2) and P(12): How many ways to make each?**
   Expected Physical Finding: **Both have only 1 way (1/36 each)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [6] [/] [36] -> 0.1667 (16.67% for Sum 7)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
6	X: 6	Permutations totaling 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1)
36	Y: 6 | X: 36	Total 6×6 Cartesian sample space
/	X: 0.1667	Peak probability for sum 7 = 1/6

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 6–8+

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: Expected Value & Game Design

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Can you ever win $3.50 on a single roll of a die?'
- Ask: 'Why is expected value an average over time rather than a possible single outcome?'
- Ask: 'How do state lotteries use expected value to guarantee millions in state revenue?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking $3.50 means you roll 3 on one turn and 4 on the next. Expected value is a weighted long-term mathematical limit.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A coin game pays $4 for heads and $0 for tails. What is the fair entry fee?"
""

OFFICIAL SOLUTION KEY:

(0.5 × $4) + (0.5 × $0) = $2.00.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/dice-and-spinner/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: Expected Value & Game Design

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What makes a game 'fair'? If a game is fair, neither the player nor the house makes money in the long run. In this lab, calculate the exact mathematical fair pri

STEP-BY-STEP APPARATUS SETUP:

1. Consider this dice game: Roll one fair 18mm die.
2. The house pays you the dollar amount showing on the die: Roll 1 -> Win $1. Roll 6 -> Win $6.
3. Question: What is a fair price to charge a player to enter this game?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Two 18mm Precision Dice + SC-32 Hardware Stack

DATUM TARGET: Expectation Formula: E(X) = Sum(x_i · P(x_i)) -> Break-even Price for Fair Games

CALCULATOR KEY: [7] [ENTER] [3.5] [*] -> E(X) Calculation

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 12, Lesson 12.5 (Expected Value)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7 (Game Theory & Fair Value)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 2, Ch. 10 • CCSS.MATH.HSS.MD.A.2 • TEKS A2.8D
GRADE LEVEL BAND:	Grades 7–9 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 9, pp. 156–164
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 6 (Expected Value Invariant) & Page 7 (RPN TVM)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
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4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: CALCULATING EXPECTED VALUE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Every outcome {1, 2, 3, 4, 5, 6} has equal probability 1/6.

Step 2: Calculate average outcome: (1 + 2 + 3 + 4 + 5 + 6) / 6 = 21 / 6 = $3.50.

Step 3: Conclude: Over 100 rolls, the player expects to win $350 ($3.50 per roll on average).

Step 4: If the host charges $3.50 to play, the game is FAIR. If the host charges $4.00, the host profits $0.50 per game!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Expectation Formula: E(X) = Sum(x_i · P(x_i)) -> Break-even Price for Fair Games

PHYSICAL APPARATUS MANIFEST:

Two 18mm Precision Dice + SC-32 Hardware Stack

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: A DOUBLE-DICE CASINO CHALLENGE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

A game pays $10 if you roll a sum of 7 on 2d6. It pays nothing for any other sum.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Probability of rolling a 7.**
   Expected Physical Finding: 6/36 = 1/6

2. **Calculate expected payout: (1/6) × $10.**
   Expected Physical Finding: $10 / 6 = $1.67

3. **If entry costs $2.00, who has the advantage and by how much?**
   Expected Physical Finding: The casino profits by $2.00 – $1.67 = $0.33 per play

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [7] [ENTER] [3.5] [*] -> E(X) Calculation

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3.5	X: 3.5	Average value of a single 6-sided die
ENTER	Y: 3.5 | X: 3.5	Stack lift
2	Y: 3.5 | X: 2	Number of dice
*	X: 7.0	Expected sum of 2d6 = 7.00

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

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- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Sample Space Trees & RPN Statistics

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it usually easier to calculate the probability of an event NOT happening, then subtract from 1?'
- Ask: 'How does the SC-32 y^x function make evaluating large sample spaces instant?'
- Ask: 'Why are casinos so certain of their profit even when players occasionally win big?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding probabilities instead of multiplying for independent events (e.g., doing 1/6 + 1/6 instead of 1/6 × 1/6). Remind them: every branch MULTIPLIES!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If you flip 10 coins, how many total possible outcomes exist in the sample space?"
""

OFFICIAL SOLUTION KEY:

2^{10} = 1,024 outcomes.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Sample Space Trees & RPN Statistics

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What are the odds of rolling 'Yahtzee' (four matching dice on a single roll)? In this lab, use sample space tree diagrams and the SC-32 power key to solve comple

STEP-BY-STEP APPARATUS SETUP:

1. Take all four precision dice from your kit.
2. Roll all 4 dice simultaneously.
3. Record whether you get: all different numbers, one pair, two pairs, three of a kind, or four of a kind.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Four 18mm Precision Dice + SC-32 Scientific Hardware Calculator

DATUM TARGET: Combinatoric Sample Space: Total Outcomes = 6^k (6^1=6, 6^2=36, 6^3=216, 6^4=1,296)

CALCULATOR KEY: [6] [ENTER] [4] [y^x] -> 1,296 (4d6 Sample Space)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 12 (Permutations & Combinations)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 7 (Culminating Probability Systems)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 2, Ch. 10 • CCSS.MATH.HSS.CP.B.9 • TEKS A2.8E
GRADE LEVEL BAND:	Grades 7–10+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 9 ('Statistics & Curve Fitting'), pp. 164–178
HARDWARE UNBOXING BOOKLET:	PROB-01 Booklet: Page 7 (RPN Statistics Connection) & Page 8 (Exercises)

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4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: EXPONENTIAL SAMPLE SPACES

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: One die: 6 outcomes. Two dice: 6 × 6 = 36 outcomes.

Step 2: Three dice: 6 × 6 × 6 = 216 outcomes. Four dice: 6^4 = 1,296 total outcomes!

Step 3: Calculate odds of rolling four 1s: Exactly 1 outcome out of 1,296 = 1/1296 (0.077%).

Step 4: Calculate odds of rolling ANY matching four of a kind: 6 outcomes (all 1s, all 2s... all 6s) / 1,296 = 6/1296 = 1/216 (0.46%).

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✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Combinatoric Sample Space: Total Outcomes = 6^k (6^1=6, 6^2=36, 6^3=216, 6^4=1,296)

PHYSICAL APPARATUS MANIFEST:

Four 18mm Precision Dice + SC-32 Scientific Hardware Calcula
tor

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 7–10+

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PROB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE BIRTHDAY PARADOX EXTENSION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

What is the probability of rolling 4 dice and having NO TWO DICE MATCH (all four different)?

FORM A MASTER STATION TASKS (Complete All 3):

1 **Die 1: 6 choices. Die 2: 5 choices. Die 3: 4 choices. Die 4: 3 choices.**
Expected Physical Finding: **6 × 5 × 4 × 3 = 360 favorable outcomes**

2 **Divide by total outcomes: 360 / 1296.**
Expected Physical Finding: **360 / 1296 = 5/18 ≈ 27.78%**

3 **What is the probability that AT LEAST TWO dice match?**
Expected Physical Finding: **Complement: 1 − 27.78% = 72.22% (Over 72% chance of a match!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[6] [ENTER] [4] [y^x] -> 1,296 (4d6 Sample Space)**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
6	X: 6	Single die outcomes
ENTER	Y: 6 | X: 6	Stack lift
4	Y: 6 | X: 4	Number of independent dice
y^x	X: 1,296	Exponential sample space volume: 6^4 = 1,296

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 7–10+

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

SC-32-HW • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Postfix Revolution: Zero Parentheses

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do ordinary calculators force you to think backwards with parentheses?'
- Ask: 'How does seeing all 4 registers on the screen keep you from losing your place?'
- Ask: 'Why did the Apollo guidance computer use RPN stack architecture?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to press an operator before typing both numbers. Remind them: 'Stack your blocks first, then e
xecute the tool!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Evaluate (15 + 5) / (7 − 2) on the SC-32 stack. Write your keystroke sequence."
""

OFFICIAL SOLUTION KEY:

[15] [ENTER] [5] [+] [7] [ENTER] [2] [−] [/] -> Result: 4.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/sc32-calculator/#product-videos

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- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

SC-32-HW • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Postfix Revolution: Zero Parentheses

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does NASA, Hewlett-Packard, and every serious engineer prefer RPN calculators? Because an RPN stack never requires parentheses and never loses intermediate

STEP-BY-STEP APPARATUS SETUP:

1. Turn on your SC-32 calculator.
2. Look at the 4-line display: notice the four stack registers labeled T (top), Z, Y, and X (bottom).
3. Notice: Register X is where you type new numbers; Register Y holds the number waiting below it.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: StackCalc SC-32 Hardware Scientific Calculator (4-Line Display)

DATUM TARGET: Postfix Invariant: Operands First, Operator Follows -> Eliminates Parentheses Completely

CALCULATOR KEY: [4] [ENTER] [3] [+] -> 7

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7A, Chapter 2 (Order of Operations & Algorithms)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 8 (Computational Algorithms)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 1, Ch. 1 • CCSS.MATH.HSA.SSE.A.1 • TEKS A.10A
GRADE LEVEL BAND:	Grades 7–12+ • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 2 ('RPN Stack Dynamics'), pp. 32–46
HARDWARE UNBOXING BOOKLET:	SC-32-HW Booklet: Page 2 (RPN Stack Invariant) & Page 5 (Tutorial 1)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

SC-32-HW • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE ZERO-PARENTHESES TEST

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: On an ordinary algebraic calculator, to compute (4 + 3) × (8 − 2), you must type 11 keystrokes and 4 parentheses.

Step 2: On SC-32 RPN: Key [4] [ENTER] [3] [+]. Register X displays 7.

Step 3: Now key [8] [ENTER] [2] [−]. Register X displays 6; Register Y displays 7!

Step 4: Now press [*]. Result: 42! Zero parentheses, zero syntax errors, zero lost numbers.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Postfix Invariant: Operands First, Operator Follows -> Eliminates Parentheses Completely

PHYSICAL APPARATUS MANIFEST:

StackCalc SC-32 Hardware Scientific Calculator (4-Line Displ
ay)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 7–12+

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

SC-32-HW • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: STACK TOWER DUALITY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Notice how RPN keystrokes mirror physical block stacking.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Place Block [4] on desk. Stack Block [3] on top.**
   Expected Physical Finding: **Physical tower of two blocks**

2. **Press [+] to combine: What happens?**
   Expected Physical Finding: **The two blocks fuse into Block [7]**

3. **Why can an RPN calculator NEVER give a 'Syntax Error'?**
   Expected Physical Finding: **Because operators only execute on existing numbers in the stack!**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [4] [ENTER] [3] [+] -> 7

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
4	X: 4	Key bottom block of tower
ENTER	Y: 4 | X: 4	Stack lift pushes 4 into register Y
3	Y: 4 | X: 3	Key top block of tower into register X
+	X: 7	Operator executes immediately, collapsing stack

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 7–12+

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

SC-32-HW • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: The 4-Register Lift & Drop

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is the stack called Last-In, First-Out (LIFO)?'
- Ask: 'How does seeing intermediate results prevent calculation errors?'
- Ask: 'What happens if you push a 5th number when the stack is full?' (Old T falls off the top).

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students pressing ENTER after every single keystroke. In RPN, you only press ENTER to separate numbers that ar
e typed consecutively!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Load 5, 4, 3, 2 into the stack. Press [*] [*] [*]. What is the final result?"
""

OFFICIAL SOLUTION KEY:

2 × 3 = 6; 6 × 4 = 24; 24 × 5 = 120 (4! factorial).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/sc32-calculator/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

SC-32-HW • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: The 4-Register Lift & Drop

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How does a 4-level memory stack move when numbers flow through it? In this lab, trace the physical lifters and gravity drops inside the SC-32 engine.

STEP-BY-STEP APPARATUS SETUP:

1. Clear the calculator stack: press [CLEAR STACK].
2. Look at the four lines on the screen: T: 0.0000, Z: 0.0000, Y: 0.0000, X: 0.0000.
3. Prepare to push four numbers onto the stack.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	SC-32 Hardware Calculator + 4 Color-Coded Register Cards (X, Y, Z, T)
DATUM TARGET:	Stack Kinematics Invariant: [ENTER] lifts X->Y->Z->T; Binary Operation drops T->Z->Y->X with T-replicate
CALCULATOR KEY:	[1] [ENTER] [2] [ENTER] [3] [ENTER] [4] -> X:4, Y:3, Z:2, T:1

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7A, Chapter 2, Lesson 2.4 (Memory & Stacks)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Computer Science & STEM, Ch. 3 • CCSS.MATH.HSA.SSE.A.1
GRADE LEVEL BAND:	Grades 7-12+ • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 2, pp. 46-58
HARDWARE UNBOXING BOOKLET:	SC-32-HW Booklet: Page 5 (Stack Lift & Drop) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

SC-32-HW • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: THE STACK LIFT CYCLE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Key [10] [ENTER]. Notice: 10 moves up into Register Y.

Step 2: Key [20] [ENTER]. Notice: 10 moves up to Z; 20 moves up to Y.

Step 3: Key [30] [ENTER]. Notice: 10 reaches top register T; 20 is in Z; 30 is in Y.

Step 4: Key [40]. Now all 4 registers hold numbers: T=10, Z=20, Y=30, X=40.

Step 5: Press [+]: Y (30) and X (40) add together to make 70! The stack drops: Z (20) drops to Y; T (10) drops to Z!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Stack Kinematics Invariant: [ENTER] lifts X->Y->Z->T; Binary Operation drops T->Z->Y->X with T-replicate

PHYSICAL APPARATUS MANIFEST:

SC-32 Hardware Calculator + 4 Color-Coded Register Cards (X, Y, Z, T)

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

SC-32-HW • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE T-REGISTER REPLICATE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

What happens to the top register T when the stack drops?

FORM A MASTER STATION TASKS (Complete All 3):

1 **Press [+] again on your result.**
Expected Physical Finding: **Y (20) and X (70) add to make 90; Z (10) drops to Y**

2 **Press [+] a third time.**
Expected Physical Finding: **Y (10) and X (90) add to make 100; T (10) drops to Y!**

3 **Notice: Register T duplicates itself on stack drop!**
Expected Physical Finding: **T-replicate allows constant-multiplier calculations**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [ENTER] [2] [ENTER] [3] [ENTER] [4] -> X:4, Y:3, Z:2, T:1

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Key 1
ENTER	Y: 1 | X: 1	Lift 1 to Y
2	Y: 1 | X: 2	Key 2
ENTER	Z: 1 | Y: 2 | X: 2	Lift 1 to Z, 2 to Y
3	Z: 1 | Y: 2 | X: 3	Key 3
ENTER	T: 1 | Z: 2 | Y: 3 | X: 3	Fill stack: 1 to T, 2 to Z, 3 to Y

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

SC-32-HW • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: Nested Expressions & Chain Calculations

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does working from the inside out match human logical reasoning?'
- Ask: 'How does the [1/x] reciprocal key eliminate the need to calculate denominators separately?'
- Ask: 'What happens on an algebraic calculator if you forget one closing parenthesis in a 4-line formula?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to write down 0.0145 on scrap paper to retype it. Emphasize: 'Never retype a number the calcul
ator already knows!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Calculate the volume of a sphere V = (4/3) * pi * r³ for r = 3.5 cm on the SC-32."
""

OFFICIAL SOLUTION KEY:

[3.5] [ENTER] [3] [y^x] [pi] [*] [4] [*] [3] [/] -> 179.59 cm³.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

SC-32-HW • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: Nested Expressions & Chain Calculations

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

In high school physics and chemistry, formulas have fractions inside fractions with square roots. In an RPN stack, you solve from the inside out with zero scrate

STEP-BY-STEP APPARATUS SETUP:

1. Consider the parallel resistance formula: R_total = 1 / (1/R1 + 1/R2).
2. Problem: Resistor 1 is 100 Ohms. Resistor 2 is 220 Ohms.
3. Calculate R_total on your SC-32.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: SC-32 Hardware Scientific Calculator

DATUM TARGET: Pipeline Invariant: Solve Innermost Parentheses First -> Automatic Register Stash

CALCULATOR KEY: [3] [ENTER] [4] [x^2] [+] [sqrt] -> 5.0000 (Pythagorean 3-4-5)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

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SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8A, Chapter 2 (Complex Order of Operations)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 2, Ch. 1 • CCSS.MATH.HSA.SSE.A.1 • TEKS A2.7A
GRADE LEVEL BAND:	Grades 8-12+ • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 2, pp. 58-68
HARDWARE UNBOXING BOOKLET:	SC-32-HW Booklet: Page 6 (Aha! Discovery: Chain Calculations)

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2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

SC-32-HW • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE RECIPROCAL PIPELINE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Key [100] [1/x]. (Register X reads 0.0100).

Step 2: Key [220] [1/x]. (Register X reads 0.0045; Register Y holds 0.0100).

Step 3: Press [+]: The two conductances add together: 0.0145.

Step 4: Press [1/x] to invert back to resistance: Result = 68.75 Ohms!

Step 5: Total keystrokes: only 10 keystrokes! Zero memory variables, zero parentheses.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

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✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Pipeline Invariant: Solve Innermost Parentheses First -> Automatic Register Stash

PHYSICAL APPARATUS MANIFEST:

SC-32 Hardware Scientific Calculator

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 8–12+

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

SC-32-HW • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THREE PARALLEL RESISTORS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Add a third resistor of 470 Ohms in parallel: R = 1 / (1/100 + 1/220 + 1/470).

FORM A MASTER STATION TASKS (Complete All 3):

1 **Key [100] [1/x] [220] [1/x] [+] [470] [1/x] [+].**
Expected Physical Finding: **Total conductance = 0.01667 S**

2 **Press [1/x].**
Expected Physical Finding: **Total resistance = 59.97 Ohms**

3 **Notice: The calculation flowed in a single continuous stream!**
Expected Physical Finding: **Zero intermediate numbers written down**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [ENTER] [4] [x^2] [+] [sqrt] -> 5.0000 (Pythagorean 3-4-5)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3	X: 3	Base side a
ENTER	Y: 3 | X: 3	Stack lift
4	Y: 3 | X: 4	Base side b
x²	Y: 3 | X: 16	Square side b: 4² = 16
x<>y	Y: 16 | X: 3	Swap registers to access side a
x²	Y: 16 | X: 9	Square side a: 3² = 9

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

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- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 8–12+

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

SC-32-HW • TENT 4/6 | ADVANCED | 20 Min

PANEL 4: Transcendental Functions: Roots & Powers

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do bank savings accounts and compound interest use e^rt?'
- Ask: 'How does the SC-32 handle numbers with 99 zeros using scientific notation (EE)?'
- Ask: 'What is the inverse function of e^x?' (Natural log ln!).

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students entering base and exponent in reverse on y^x. In RPN: Y is the base (typed first), X is the exponent (typed second).

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Calculate the pH of a solution with hydrogen ion concentration [H+] = 3.2 × 10^-5 M. (pH = -log[H+])."
""

OFFICIAL SOLUTION KEY:

[3.2] [EE] [5] [+/-] [log] [+/-] -> pH = 4.49 (Acidic).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

SC-32-HW • TENT 4/6 | ADVANCED | 20 Min

PANEL 1: Transcendental Functions: Roots & Powers

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

From earthquake Richter scales to bacterial colony growth, nature works on exponential and logarithmic curves. In this lab, use the SC-32 transcendental engine t

STEP-BY-STEP APPARATUS SETUP:

1. Look at the transcendental key block on the SC-32: [y^x], [sqrt], [e^x], [ln], [10^x], [log].
2. Scenario: A bacterial culture of 500 cells doubles every 3 hours.
3. Formula: N(t) = 500 × 2^(t/3). How many cells exist after 12 hours?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: SC-32 Hardware Scientific Calculator

DATUM TARGET: Power Invariant: y^x evaluates exponential growth; ln and e^x evaluate continuous natural systems

CALCULATOR KEY: [10] [ENTER] [6] [y^x] -> 1,000,000 (10^6)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8A, Chapter 1 (Exponents & Scientific Notation)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 2, Ch. 6 • CCSS.MATH.HSF.LE.A.4 • TEKS A2.5B
GRADE LEVEL BAND:	Grades 9-12+ • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 3 ('Algebra & Transcendentals'), pp. 68-78
HARDWARE UNBOXING BOOKLET:	SC-32-HW Booklet: Page 5 (Transcendental Keys) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

SC-32-HW • TENT 4/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: EXPONENTIAL TIME MODELING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Calculate the doubling periods: 12 hours / 3 hours = 4 doublings.

Step 2: Key base: [2] [ENTER] [4] [y^x]. (Register X reads 16).

Step 3: Multiply by initial population: [500] [*]. Result = 8,000 cells!

Step 4: Now test non-integer hours: How many cells after 7.5 hours? [7.5] [ENTER] [3] [/] (2.5 periods). Then [2] [x<>y] [y^x] [500] [*] = 2,828 cells.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Power Invariant: y^x evaluates exponential growth; ln and e^x evaluate continuous natural systems

PHYSICAL APPARATUS MANIFEST:

SC-32 Hardware Scientific Calculator

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

SC-32-HW • TENT 4/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: SOLVING FOR TIME WITH LOGARITHMS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

When will the population reach 50,000 cells? 500 × 2^(t/3) = 50,000.

FORM A MASTER STATION TASKS (Complete All 3):

1 **Divide both sides by 500: 50,000 / 500.**
Expected Physical Finding: **100 (Culture has grown 100-fold)**

2 **Take natural log of 100: [100] [ln].**
Expected Physical Finding: **4.6052**

3 **Divide by ln(2): [2] [ln] [/].**
Expected Physical Finding: **6.6439 doubling periods**

4 **Multiply by 3 hours per period: [3] [*].**
Expected Physical Finding: **t = 19.93 hours (Under 20 hours!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [10] [ENTER] [6] [y^x] -> 1,000,000 (10^6)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
10	X: 10	Base
ENTER	Y: 10 | X: 10	Stack lift
6	Y: 10 | X: 6	Power
y^x	X: 1,000,000	Evaluate power

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

SC-32-HW • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: The Rectangular–Polar Engine

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is ->POL faster than doing sqrt(x² + y²) and arctan(y/x) by hand?'
- Ask: 'How does ->POL handle negative coordinates like (−30, −40) without quadrant confusion?'
- Ask: 'How does the SC-32 connect the physical Volvelle tool to digital electronic engineering?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students in RAD mode expecting degree answers. Always check the top status annunciator: [DEG] for degrees, [RAD] for radians!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A drone flies at 120 km/h at heading 135°. What are its North and East velocity components?"
""

OFFICIAL SOLUTION KEY:

[120] [ENTER] [135] [->REC] -> East = −84.85 km/h (West), North = +84.85 km/h.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/sc32-calculator/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

SC-32-HW • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: The Rectangular–Polar Engine

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Whether designing antennas, analyzing AC circuits, or steering an aircraft in crosswinds, engineers constantly convert between (X, Y) and (r, theta). The SC-32 (

STEP-BY-STEP APPARATUS SETUP:

1. Set the SC-32 angle mode to [DEG] (Degrees).
2. Problem: An offshore wind turbine is located 60 km East and 80 km North of the harbor.
3. Find the direct distance (r) and compass bearing (theta) to the turbine.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	SC-32 Hardware Scientific Calculator + Unit Circle Volvelle
DATUM TARGET:	Coordinate Invariant: (x, y) <-> (r, theta) via r = sqrt(x²+y²), theta = atan2(y, x)
CALCULATOR KEY:	[3] [ENTER] [4] [->POL] -> r: 5.0000, theta: 53.13°

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math, Ch. 10 (Polar Coordinates)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 6, Sec. 6.4 • CCSS.MATH.HSN.VM.A.3 • TEKS P.4H
GRADE LEVEL BAND:	Grades 10–12+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4 ('Vectors & Polar'), pp. 118–130
HARDWARE UNBOXING BOOKLET:	SC-32-HW Booklet: Page 6 (Vector Conversions) & Page 7 (Ecosystem Connection)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

SC-32-HW • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE ONE-TOUCH POLAR CONVERSION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Key horizontal X: [60] [ENTER].

Step 2: Key vertical Y: [80]. (Notice: Y holds 60, X holds 80).

Step 3: Press [->POL]: Instantly, Register X displays the straight-line distance: 100.00 km!

Step 4: Press [x<>y] to view Register Y: Compass angle = 53.13° North of East!

Step 5: Verify on the Volvelle: Align rotor to 53.1°: Notice X ≈ 0.60 and Y ≈ 0.80 (multiplied by 100 = 60 and 80!).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Coordinate Invariant: (x, y) <-> (r, theta) via r = sqrt(x²+y²), theta = atan2(y, x)

PHYSICAL APPARATUS MANIFEST:

SC-32 Hardware Scientific Calculator + Unit Circle Volvelle

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

SC-32-HW • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: COMPLEX IMPEDANCE IN AC CIRCUITS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

An electrical circuit has resistance R = 50 Ohms and inductive reactance X_L = 35 Ohms (Z = 50 + j35).

FORM A MASTER STATION TASKS (Complete All 3):

1 **Key [50] [ENTER] [35] [->POL].**
Expected Physical Finding: **Register X: Impedance |Z| = 61.03 Ohms**

2 **Check phase angle in Register Y: [x<>y].**
Expected Physical Finding: **Phase angle theta = 34.99°**

3 **Convert back to rectangular: [->REC].**
Expected Physical Finding: **Returns exactly to X: 50.00, Y: 35.00**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [ENTER] [4] [->POL] -> r: 5.0000, theta: 53.13°

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3	X: 3	Horizontal X component
ENTER	Y: 3 | X: 3	Stack lift
4	Y: 3 | X: 4	Vertical Y component
->POL	Y: 53.13 | X: 5.0000	Convert Rectangular (3,4) to Polar (r=5, theta=53.13°)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

SC-32-HW • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Algorithmic Efficiency & Speed Mastery

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does reducing keystrokes by 30% dramatically reduce typo errors?'
- Ask: 'How does tactile dome feedback (400g snap) ensure your finger knows a key registered?'
- Ask: 'How does the SC-32 hardware calculator complete the entire 9-kit StackCalc Learning Lab?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Congratulations! You have mastered the entire StackCalc connected math ecosystem from K-2 tactile Ten-Frames to professi
onal RPN scientific calculation!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Solve the UIL contest sprint problem: [sqrt(2.54² + 8.12²) − 3.45] / 1.78 on SC-32."
""

OFFICIAL SOLUTION KEY:

[2.54] [x²] [8.12] [x²] [+] [sqrt] [3.45] [−] [1.78] [/] -> 2.8427.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/sc32-calculator/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

SC-32-HW • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Algorithmic Efficiency & Speed Mastery

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

In high-stakes math competitions (UIL, Math Olympiad, AMC) and engineering exams, every second counts. In this capstone lab, race the SC-32 RPN against any stand

STEP-BY-STEP APPARATUS SETUP:

1. Team up with a partner using an algebraic calculator (TI-30, TI-84, or Casio).
2. Both partners will solve the exact same 5 competition math problems.
3. Partner A uses SC-32 RPN; Partner B uses the algebraic calculator. Count every keystroke!

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	SC-32 Hardware Calculator + Benchmark Stopwatch + Contest Problem Sheet
DATUM TARGET:	Keystroke Efficiency Invariant: RPN Requires 20-35% Fewer Keystrokes with ZERO Parenthesis Overhead
CALCULATOR KEY:	[5] [ENTER] [3] [+] [8] [ENTER] [2] [−] [*] -> 48 (10 Keystrokes vs 14 on TI)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 14 (Computational Algorithms)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Secondary STEM, Ch. 12 • UIL & Contest Mathematics Standards
GRADE LEVEL BAND:	Grades 9-12+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 14 ('UIL Contest Playbook & Speed Drills'), pp. 240-256
HARDWARE UNBOXING BOOKLET:	SC-32-HW Booklet: Page 7 (RPN Postfix Advantage) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

SC-32-HW • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE 5-PROBLEM SPEED BENCHMARK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Problem 1: (14 × 23) + (18 × 9). SC-32: [14] [ENTER] [23] [*] [18] [ENTER] [9] [*] [+] (11 keys). TI: [(] 14 [*] 23 [)] [+] [(] 18 [*] 9 [)] [=] (15 keys).

Step 2: Problem 2: sqrt(12² + 35²). SC-32: [12] [x²] [35] [x²] [+] [sqrt] (6 keys). TI requires opening brackets and equals (9 keys).

Step 3: Problem 3: 1 / (1/24 + 1/36). SC-32: [24] [1/x] [36] [1/x] [+] [1/x] (6 keys).

Step 4: Tally the scores: SC-32 uses 28% fewer keystrokes and completes the benchmark in half the time!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Keystroke Efficiency Invariant: RPN Requires 20–35% Fewer Keystrokes with ZERO Parenthesis Overhead

PHYSICAL APPARATUS MANIFEST:

SC-32 Hardware Calculator + Benchmark Stopwatch + Contest Problem Sheet

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

SC-32-HW • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: ERROR RECOVERY AUDIT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

What happens when you mistype a single number in a 20-step calculation?

FORM A MASTER STATION TASKS (Complete All 3):

1. **On an algebraic calculator with a long line of text:**
   Expected Physical Finding: **Must scroll cursor back through line or re-enter entire equation**

2. **On SC-32 RPN:**
   Expected Physical Finding: **Simply press [DROP] or [x<>y] to fix the top register and continue immediately!**

3. **Why do competitive UIL math champions use RPN?**
   Expected Physical Finding: **Fewer keystrokes, instant error correction, zero parentheses.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [5] [ENTER] [3] [+] [8] [ENTER] [2] [−] [*] -> 48 (10 Keystrokes vs 14 on TI)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
5+3	X: 8	Evaluate first subterm
8−2	Y: 8 | X: 6	Evaluate second subterm
*	X: 48	Multiply in single stroke: 48

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The 12:1 Epicyclic Reduction

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do ordinary toy clocks slip and break when children turn them backwards?'
- Ask: 'How does the planetary gear core prevent fingers from getting pinched?'
- Ask: 'Why is there 360 degrees in a full circle and 30 degrees between each hour?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students trying to force the hour pointer directly with their thumbs. Direct them to drive the clock smoothly using the minute hand drive shaft.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"If the minute hand turns through 90° (15 minutes), how many degrees does the hour hand move?"
""

OFFICIAL SOLUTION KEY:

Hour hand moves 30° / 4 = 7.5°.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The 12:1 Epicyclic Reduction

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does turning the long hand move the short hand automatically? Inside this clock is a miniature planetary gearbox identical to the transmissions in spacecraf

STEP-BY-STEP APPARATUS SETUP:

1. Hold the clock body in your hands.
2. Align both the hour pointer and minute pointer to 12:00.
3. Slowly rotate the minute hand one full revolution clockwise.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core (Herringbone Gearbox) + Detachable Dial Face

DATUM TARGET: Gear Invariant: 1 Full Turn of Minute Hand (360°) -> Exactly 30° Hour Hand Advance

CALCULATOR KEY: [CLK] [1] [:] [00] [+] [1] [:] [00] -> 2:00

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 1B, Chapter 15 (Telling Time to the Hour)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 1, Unit 4 (Clocks & Hours)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 1, Ch. 9 • CCSS.MATH.1.MD.B.3 • TEKS 1.7E

GRADE LEVEL BAND: Grades 1–3 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 6 ('Geometry & Time'), pp. 128–140

HARDWARE UNBOXING BOOKLET: TIME-01 Booklet: Page 2 (12:1 Epicyclic Ratio) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 12:1 REDUCTION CHECK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Watch the hour pointer as the minute hand completes 1 full turn (360°).

Step 2: Notice: The hour hand advances from 12 to 1 (exactly 30° arc, which is 1/12 of 360°).

Step 3: Rotate the minute hand another full turn. Notice: Hour pointer moves exactly from 1 to 2.

Step 4: Conclude: 1 hour = 60 minutes. The 12:1 gear ratio guarantees that the hands can NEVER slip out of synchronization!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Gear Invariant: 1 Full Turn of Minute Hand (360°) -> Exactly 30° Hour Hand Advance

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core (Herringbone Gearbox) + Detachable Dial Face

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: REVERSE ROTATION AUDIT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Rotate the minute hand counter-clockwise by 2 full turns.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What time does the clock display after 2 reverse turns from 2:00?**
   Expected Physical Finding: **12:00**

2. **Did the gears slip or jam during reverse motion?**
   Expected Physical Finding: **Zero slip (Herringbone captive teeth)**

3. **How many minute turns are needed to move the hour hand 180° (6 hours)?**
   Expected Physical Finding: **6 full revolutions**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [CLK] [1] [:] [00] [+] [1] [:] [00] -> 2:00

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1:00	X: 1:00	Initial time state
ENTER	Y: 1:00 | X: 1:00	Stack lift
1:00	Y: 1:00 | X: 1:00	Interval duration
+	X: 2:00	Time addition evaluates new hour

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: Halfway There: Continuous Motion

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Where would the hour hand point at 8:30?'
- Ask: 'Can the hour pointer ever point directly at a number when the minute hand is at 6?'
- Ask: 'How does the gear drive keep you from being fooled on tests?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students reading 5:50 as 6:50. Have them trace the arc backwards to 5:00 to see that the hour hand has not cro
ssed the 6:00 finish line.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"At what exact time between 1:00 and 2:00 are the hour and minute hands pointing in EXACTLY opposite directions (180"
"°)?"

OFFICIAL SOLUTION KEY:

At approximately 1:38 and 11 seconds (1:38.18).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: Halfway There: Continuous Motion

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

The #1 mistake kids make on tests is thinking that at 3:30, the hour hand still points directly at 3. Watch how physical kinematics kills this mistake forever.

STEP-BY-STEP APPARATUS SETUP:

1. Set the clock to 3:00 (minute hand at 12, hour pointer at 3).
2. Slowly turn the minute hand half a turn until it points straight down at 6 (30 minutes).
3. Look closely at the red hour pointer: Where is it pointing?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core + Detachable Half-Hour Overlay Disc

DATUM TARGET: Kinematic Invariant: At :30 minutes, Hour Pointer is EXACTLY 15° (midway between numbers)

CALCULATOR KEY: [3] [:] [30] -> Display: 3:30 (Hour hand at 3.5)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2B, Chapter 10 (Telling Time to 5 Minutes)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 5 (Clocks & Half-Hours)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 12 • CCSS.MATH.2.MD.C.7 • TEKS 2.9G
GRADE LEVEL BAND:	Grades 1–3 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 132–136
HARDWARE UNBOXING BOOKLET:	TIME-01 Booklet: Page 2 (Continuous Motion) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE MIDWAY HORIZON

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Notice: The hour pointer has crawled forward. It does NOT point at 3!

Step 2: Lay a straightedge from the center arbor through the hour pointer: It points EXACTLY halfway between 3 and 4 (at the 15° tick).

Step 3: Now advance to 3:45 (minute hand at 9): Notice the hour pointer is 3/4 of the way to 4.

Step 4: Conclude: The hour hand NEVER jumps suddenly at the bell—it glides continuously at 1/12 the speed of the minute hand.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Kinematic Invariant: At :30 minutes, Hour Pointer is EXACTLY 15° (midway between numbers)

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Detachable Half-Hour Overlay Disc

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE 5:50 NEAR-MISS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Set the clock to 5:50 (minute hand at 10). Where is the hour pointer?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Is the hour pointer closer to 5 or closer to 6?**
   Expected Physical Finding: **Much closer to 6 (it has traveled 50/60 = 83% of the way)**

2. **Why do many students mistakenly read 5:50 as 6:50?**
   Expected Physical Finding: **Because the hour hand looks like it is pointing at 6**

3. **How do you know it is still 5:00 and not 6:00?**
   Expected Physical Finding: **The minute hand has not yet reached 12**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [:] [30] -> Display: 3:30 (Hour hand at 3.5)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3:00	X: 3:00	Base hour
30m	Y: 3:00 | X: 0:30	Elapsed interval
+	X: 3:30	Hour hand position = 3 + 30/60 = 3.5

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 4: Concentric Track Reading

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why are there 12 numbers on the clock if there are 60 minutes in an hour?'
- Ask: 'How does multiplying by 5 help you read the minute hand faster?'
- Ask: 'What track do you look at to read 4:08 vs 4:40?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students confusing 4:08 with 4:40. Point out that the minute hand points at 8 for 4:40, but at 1 + 3 ticks for 4:08.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A clock shows 9:47. How many minutes remain until 10:00?"
""

OFFICIAL SOLUTION KEY:

60 − 47 = 13 minutes.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 1: Concentric Track Reading

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How does a clock dial pack three different measuring tracks into a single circular face? In this lab, you master the 3 concentric reading tracks.

STEP-BY-STEP APPARATUS SETUP:

1. Inspect the detachable dial face.
2. Trace the three concentric circular paths from outside to inside.
3. Track 1 (Outer): 60 micro-ticks (1 minute each). Track 2 (Middle): 12 five-minute intervals. Track 3 (Inner): 12 hour numerals.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Planetary Clock Core + Standard 3-Track Detachable Dial Face
DATUM TARGET:	Concentric Geometry: Outer Rim = 60 Ticks (1 min) | Mid = 12 Intervals (5 min) | Inner = 12 Hours
CALCULATOR KEY:	[5] [ENTER] [7] [*] -> 35 Min (Hand at 7)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2B, Chapter 10, Lesson 10.2 (Minutes Past the Hour)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 5, Module 2 (Skip Counting Minutes by 5s)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 12, Sec. 12.2 • CCSS.MATH.2.MD.C.7 • TEKS 2.9G
GRADE LEVEL BAND:	Grades 2–4 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 136–140
HARDWARE UNBOXING BOOKLET:	TIME-01 Booklet: Page 5 (Dial Features & Tracks)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 2: LAB INVESTIGATION: THE 5-MINUTE SKIP COUNT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Point the minute hand at numeral 1: Read the middle track: '5 min'.

Step 2: Advance to 2: '10 min'. Advance to 3: '15 min'.

Step 3: Skip count by 5s around the dial: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60.

Step 4: Point the minute hand at 8: Calculate 8 × 5 = 40 minutes. Confirm with the outer tick marks.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Concentric Geometry: Outer Rim = 60 Ticks (1 min) | Mid = 12 Intervals (5 min) | Inner = 12 Hours

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Standard 3-Track Detachable Dial Face

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 2–4

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: READING TO THE EXACT MINUTE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Set the clock to 7:23. Use all three concentric tracks.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Where does the hour pointer sit?**
   Expected Physical Finding: **Between 7 and 8 (about 1/3 of the way)**

2. **How do you find 23 on the minute track?**
   Expected Physical Finding: **Find 4 (20 min) on mid track, then count 3 ticks past it on outer track**

3. **Write the time in standard digital format.**
   Expected Physical Finding: **7:23**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [5] [ENTER] [7] [*] -> 35 Min (Hand at 7)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
7	X: 7	Hour numeral on dial
5	Y: 7 | X: 5	Minutes per dial sector
*	X: 35	Calculate minute count via skip-counting

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 2–4

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: Elapsed Time as Angular Arcs

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can\'t you just subtract 2:45 – 1:15 like ordinary decimals?'
- Ask: 'How does breaking elapsed time into landmark hours (like 11:00 or 12:00) make it easy?'
- Ask: 'How many 15-minute sector blocks fit around the entire clock face?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students subtracting 10:45 from 12:15 using base-10 borrowing (getting 1.70 instead of 1 hr 30 min). Remind th
em: time resets at 60, not 100!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Flight 412 departs at 8:50 PM and lands at 11:25 PM. What is the flight duration?"
""

OFFICIAL SOLUTION KEY:

8:50 to 9:00 = 10 min. 9:00 to 11:00 = 2 hr. 11:00 to 11:25 = 25 min. Total = 2 hr 35 min.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

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- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: Elapsed Time as Angular Arcs

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How long did soccer practice last? Instead of doing confusing base-60 math in your head, measure elapsed time as a physical angular wedge.

STEP-BY-STEP APPARATUS SETUP:

1. Dock the clock stand in the center of the table.
2. Problem: A movie starts at 1:15 and ends at 2:45. How long was the movie?
3. Set the clock to 1:15.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Clock Core + 15-Minute Arc Blocks (90° sectors) + Digital Practice Lid

DATUM TARGET: Angle-Time Invariant: 90° Arc = 15 Min | 180° Arc = 30 Min | 360° Arc = 60 Min

CALCULATOR KEY: [2] [:] [45] [ENTER] [1] [:] [15] [−] -> 1:30 (1 hr 30 min)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3A, Chapter 7 (Elapsed Time)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 3, Unit 5 (Time & Measurement Arcs)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 3, Ch. 12 • CCSS.MATH.3.MD.A.1 • TEKS 3.7C
GRADE LEVEL BAND:	Grades 3-5 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 140-146
HARDWARE UNBOXING BOOKLET:	TIME-01 Booklet: Page 6 (Aha! Discovery: Elapsed Arc Sums)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE SECTOR WEDGE FIT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Advance the clock from 1:15 to 2:15 (one full 360° revolution of minute hand). Record: 1 hour.

Step 2: Now advance from 2:15 to 2:45: The minute hand sweeps from 3 to 9 (a 180° semicircle arc).

Step 3: Fit two 15-minute sector blocks into this 180° gap: Exactly two 15-min blocks fit (15 + 15 = 30 minutes).

Step 4: Sum the components: 1 hour + 30 minutes = 1 hour 30 minutes (90 minutes total elapsed time).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Angle-Time Invariant: 90° Arc = 15 Min | 180° Arc = 30 Min | 360° Arc = 60 Min

PHYSICAL APPARATUS MANIFEST:

Clock Core + 15-Minute Arc Blocks (90° sectors) + Digital Practice Lid

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: BRIDGING THE HOUR BOUNDARY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

A science lab begins at 10:45 AM and ends at 12:15 PM. Find the elapsed time.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Minutes from 10:45 to 11:00 (top of the hour).**
   Expected Physical Finding: **15 minutes (one 15-min block)**

2. **Hours from 11:00 to 12:00.**
   Expected Physical Finding: **1 hour (60 minutes)**

3. **Minutes from 12:00 to 12:15.**
   Expected Physical Finding: **15 minutes**

4. **Total elapsed time.**
   Expected Physical Finding: **1 hour 30 minutes**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [:] [45] [ENTER] [1] [:] [15] [−] -> 1:30 (1 hr 30 min)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
2:45	X: 2:45	End time
ENTER	Y: 2:45 | X: 2:45	Stack lift
1:15	Y: 2:45 | X: 1:15	Start time
−	X: 1:30	Elapsed interval difference: 90 minutes

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 4: The 24-Hour Solar Meridian

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is 24-hour time safer for hospitals and airlines than saying 8 o\'clock?'
- Ask: 'Why don\'t morning AM hours need 12 added to them?'
- Ask: 'What does AM (Ante Meridiem) and PM (Post Meridiem) mean in Latin?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding 12 to morning hours (e.g., converting 8:00 AM to 20:00). Remind them: morning is within the fi
rst 12 hours of the day!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A train leaves Paris at 22:35 and arrives in Berlin at 06:15 the next morning. How long was the journey?"
""

OFFICIAL SOLUTION KEY:

22:35 to 24:00 = 1 hr 25 min. 00:00 to 06:15 = 6 hr 15 min. Total = 7 hr 40 min.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 1: The 24-Hour Solar Meridian

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does the day reset at midnight, and why do pilots, astronauts, and doctors use 24-hour time? In this lab, flip your dial to master 24-hour military time.

STEP-BY-STEP APPARATUS SETUP:

1. Detach the standard 12-hour dial face from the clock body.
2. Flip the dial to Side B (the 24-Hour Solar Meridian Face: numbers 13 to 24 on the outer rim).
3. Snap the dial back onto the octagonal MultiBuild mounting posts.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core + Reversible 24-Hour Military Dial Face

DATUM TARGET: Modular Arithmetic: 1:00 PM = 1 + 12 = 13:00 | 11:00 PM = 11 + 12 = 23:00

CALCULATOR KEY: [13] [:] [00] [12/24] -> 1:00 PM

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 3B, Chapter 12 (24-Hour Time)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 4, Unit 5 (Solar Cycles & Navigation Time)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 4, Ch. 12 • CCSS.MATH.4.MD.A.2 • TEKS 4.8C

GRADE LEVEL BAND: Grades 3-5 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 146-152

HARDWARE UNBOXING BOOKLET: TIME-01 Booklet: Page 5 (Detachable Dual-Sided Dials)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: THE NOON CONVERSION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Set the clock to 12:00 Noon (sun at highest point in the sky).

Step 2: Advance the clock 1 hour into the afternoon: Notice the inner track reads '1', but the outer 24-hour track reads '13' (13:00).

Step 3: Advance to 6:00 PM dinner: Inner track reads '6', outer track reads '18' (18:00).

Step 4: Rule: For any PM time, add 12 to the hour: 1 + 12 = 13; 6 + 12 = 18; 11 + 12 = 23.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Modular Arithmetic: 1:00 PM = 1 + 12 = 13:00 | 11:00 PM = 11 + 12 = 23:00

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Reversible 24-Hour Military Dial Face

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: CONVERTING FLIGHT ITINERARIES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Convert between standard AM/PM time and 24-hour international military time.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Convert 3:45 PM to 24-hour time.**
   Expected Physical Finding: **3 + 12 = 15 -> 15:45**

2. **Convert 21:15 to standard AM/PM time.**
   Expected Physical Finding: **21 − 12 = 9 -> 9:15 PM**

3. **What is 12:00 Midnight in 24-hour time?**
   Expected Physical Finding: **00:00 (or 24:00)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[13] [:] [00] [12/24] -> 1:00 PM**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
13:00	X: 13:00	Military 24-hour time
12/24	X: 1:00 PM	Toggle 12-hour AM/PM meridian display

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Digital-to-Analog Kinematic Bridge

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'How does an analog clock show you how much time is LEFT in the hour better than digital numbers?'
- Ask: 'Why does a digital clock make it harder to visualize fractions like a quarter-to or half-past?'
- Ask: 'How does the RPN calculator handle the base-60 turnover at 60 minutes?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to add 7:40 + 0:25 and getting 7:65 on ordinary calculators. The SC-6 CLK mode automatically c
onverts 65 minutes into +1 hour 5 minutes (8:05).

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A timer counts down from 45:00 minutes. After 28 minutes and 40 seconds, what time remains?"
""

OFFICIAL SOLUTION KEY:

45:00 − 28:40 = 16 minutes 20 seconds.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Digital-to-Analog Kinematic Bridge

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why do modern smartwatches let you toggle between analog hands and digital numbers? In this culminating lab, you connect physical hand angles to digital calcula

STEP-BY-STEP APPARATUS SETUP:

1. Snap the Digital Practice Lid onto your Gridfinity clock box.
2. Slide the privacy shutter CLOSED over the digital display window.
3. Partner A sets a secret time on the planetary clock hands, then passes it to Partner B.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core + Digital Practice Lid with Sliding Shutter

DATUM TARGET: Bimodal Duality: Physical Angular Displacement (Analog) = Numerical State Vector (Digital)

CALCULATOR KEY: [CLK] [8] [:] [45] [ENTER] [30] [+] -> 9:15

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 3B, Chapter 12, Lesson 12.3 (Time Word Problems)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 4, Unit 5 (Digital vs Analog Time Conversion)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 4, Ch. 12 • CCSS.MATH.4.MD.A.2 • TEKS 4.8C

GRADE LEVEL BAND: Grades 2-5 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 152-158

HARDWARE UNBOXING BOOKLET: TIME-01 Booklet: Page 7 (RPN Stack Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE REVERSE ENCODING GAME

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Partner B inspects the physical clock hands and announces the time aloud.

Step 2: Partner B keys the time into the SC-6 calculator in CLK mode: e.g. [4] [:] [2] [5].

Step 3: Slide open the privacy shutter on the digital practice lid to verify the answer!

Step 4: Swap roles: Partner B sets the clock, Partner A reads and verifies.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Bimodal Duality: Physical Angular Displacement (Analog) = Numerical State Vector (Digital)

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Digital Practice Lid with Sliding Shu
tter

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE MULTI-STEP TIMELINE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

A student wakes up at 7:15 AM. Breakfast takes 25 min. Walking to school takes 15 min. When do they arrive?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Key start time into SC-6: [7] [:] [15] [ENTER].**
   Expected Physical Finding: **Y: 7:15 | X: 7:15**
2. **Add breakfast: [25] [+].**
   Expected Physical Finding: **X: 7:40**
3. **Add walk: [15] [+].**
   Expected Physical Finding: **X: 7:55 AM**
4. **Verify on the physical clock: Do the hands match 7:55?**
   Expected Physical Finding: **Minute hand at 11, hour pointer near 8**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [CLK] [8] [:] [45] [ENTER] [30] [+] -> 9:15

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
8:45	X: 8:45	Set initial analog clock time
ENTER	Y: 8:45 | X: 8:45	Stack lift
30	Y: 8:45 | X: 0:30	Key 30 minute delay
+	X: 9:15	Display updated time in register X

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: Degrees to Radians: The Natural Angle

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is 360 an arbitrary number invented by the ancient Babylonians?'
- Ask: 'Why is radian measure called natural mathematics?'
- Ask: 'How does walking around a 1-meter radius track relate to radians?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking pi radians means 3.14 degrees. Emphasize: pi radians is 180 DEGREES!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Convert 210° to exact radians using the Volvelle."
""

OFFICIAL SOLUTION KEY:

210° = 7 × 30° = 7pi/6 rad.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: Degrees to Radians: The Natural Angle

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why do mathematicians and engineers abandon 360 degrees and use radians instead? Because in a unit circle (r = 1), radians ARE the physical distance you walk ar

STEP-BY-STEP APPARATUS SETUP:

1. Hold the Unit Circle Volvelle in your hands.
2. Rotate the upper cursor rotor until the hairline aligns with 0° (Positive X-axis).
3. Look into the reading window at 0°: notice '0 rad' and coordinate (1, 0).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Precision Dual-Rotor Volvelle Slide Calculator with Exact Radical Windows
DATUM TARGET:	Radian Invariant: Arc Length s = r · theta -> When r = 1, Angle in Radians EXACTLY Equals Arc Length
CALCULATOR KEY:	[180] [->RAD] -> 3.14159... (pi radians)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math (A-Math), Ch. 8 (Radian Measure)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 4, Sec. 4.1 • CCSS.MATH.HSF.TF.A.1 • TEKS P.4A
GRADE LEVEL BAND:	Grades 9-11 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4 ('Trigonometry, Vectors & Polar'), pp. 74-86
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 2 (Polar Invariant) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE RADIAN CONVERSION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Rotate the rotor to 90° (top of circle): Read the exact radical window: 'pi/2 rad'.

Step 2: Rotate to 180°: Read 'pi rad'. Notice: Halfway around a circle is exactly pi (3.14159...).

Step 3: Rotate to 270°: Read '3pi/2 rad'. Rotate full circle to 360°: Read '2pi rad' (6.28318...).

Step 4: Align to landmark angles: 30° = pi/6; 45° = pi/4; 60° = pi/3. Notice: 30° goes into 180° exactly 6 times!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Radian Invariant: Arc Length s = r · theta -> When r = 1, Angle in Radians EXACTLY Equals Arc Length

PHYSICAL APPARATUS MANIFEST:

Precision Dual-Rotor Volvelle Slide Calculator with Exact Ra
dical Windows

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: FINDING SPECIAL ANGLES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Align the rotor to 120°, 135°, and 150° in Quadrant II.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is 120° in exact radians?**
   Expected Physical Finding: 2pi/3 rad

2. **What is 135° in exact radians?**
   Expected Physical Finding: 3pi/4 rad

3. **What is 150° in exact radians?**
   Expected Physical Finding: 5pi/6 rad

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [180] [->RAD] -> 3.14159... (pi radians)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
180	X: 180	Angle in degrees
->RAD	X: 3.14159	Evaluate exact radian equivalent (pi rad)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: Cartesian Projections: Cosine & Sine

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can cosine and sine never be greater than 1 on a unit circle?'
- Ask: 'What happens to the shadows at 180°?'
- Ask: 'How does the Volvelle eliminate the need to memorize the unit circle table?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students mixing up X and Y. Cosine is ALWAYS horizontal X (like a shadow on the floor); Sine is ALWAYS vertica
l Y (like height on the wall).

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"What are the coordinates at 240° in Quadrant III?"
""

OFFICIAL SOLUTION KEY:

(−1/2, −sqrt(3)/2) — both coordinates are negative!

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: Cartesian Projections: Cosine & Sine

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What do cosine and sine actually measure? They are nothing more than the horizontal shadow (X) and vertical shadow (Y) of a rotating clock hand of length 1.

STEP-BY-STEP APPARATUS SETUP:

1. Align the Volvelle rotor to 30°.
2. Look at the coordinate window: notice (sqrt(3)/2, 1/2).
3. Inspect the physical grid: Notice the horizontal X coordinate reaches ~0.866 and vertical Y reaches exactly 0.500.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Dual-Rotor Volvelle with Exact Coordinate Readout Windows (cos theta, sin theta)
DATUM TARGET:	Coordinate Invariant: Any Point on Unit Circle P(theta) = (cos theta, sin theta)
CALCULATOR KEY:	[30] [COS] -> 0.8660 (sqrt(3)/2) | [30] [SIN] -> 0.5000 (1/2)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math, Ch. 9 (Trigonometric Functions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 4, Sec. 4.2 • CCSS.MATH.HSF.TF.A.2 • TEKS P.4B
GRADE LEVEL BAND:	Grades 10–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 86–94
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 5 (Reading Coordinates) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: THE 45° AND 60° INVERSIONS

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Rotate to 45°: Read coordinate (sqrt(2)/2, sqrt(2)/2) = (0.707, 0.707). Both shadows are equal!

Step 2: Rotate to 60°: Read coordinate (1/2, sqrt(3)/2) = (0.500, 0.866).

Step 3: Compare 30° and 60°: Notice the X and Y coordinates SWAPPED! 30° has X=sqrt(3)/2, Y=1/2; 60° has X=1/2, Y=sqrt(3)/2.

Step 4: Rotate to 90°: X shadow collapses to 0; Y shadow reaches maximum 1. (cos 90° = 0, sin 90° = 1).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Coordinate Invariant: Any Point on Unit Circle P(theta) = (cos theta, sin theta)

PHYSICAL APPARATUS MANIFEST:

Dual-Rotor Volvelle with Exact Coordinate Readout Windows (cos theta, sin theta)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE SIGN REVERSAL IN QUADRANT II

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Rotate the rotor to 150° in Quadrant II (top-left).

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is the sign of the X coordinate (cosine) in Quadrant II?**
   Expected Physical Finding: **Negative (-sqrt(3)/2 ≈ -0.866)**

2. **What is the sign of the Y coordinate (sine) in Quadrant II?**
   Expected Physical Finding: **Positive (+1/2 = 0.500)**

3. **Why is cosine negative in Quadrant II?**
   Expected Physical Finding: **Because the point sits to the LEFT of the vertical Y-axis!**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [30] [COS] -> 0.8660 (sqrt(3)/2) | [30] [SIN] -> 0.5000 (1/2)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
30	X: 30	Angle 30 degrees
COS	X: 0.8660	Horizontal coordinate = sqrt(3)/2
30	Y: 0.8660 | X: 30	Angle
SIN	X: 0.5000	Vertical coordinate = 1/2

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: The Pythagorean Invariant

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Can the sum of squares ever equal 1.05 or 0.98?'
- Ask: 'How does this identity protect GPS satellites from calculation drift?'
- Ask: 'If cos theta = 0.6, how can you find sin theta in 5 seconds without a calculator?' (sqrt(1 − 0.36) = 0.8!).

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students writing cos(theta)² instead of (cos theta)². Clarify that the notation cos² theta means you compute t
he cosine first, then square the result.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If sin theta = 5/13 in Quadrant I, what is cos theta?"
""

OFFICIAL SOLUTION KEY:

cos theta = sqrt(1 − (5/13)²) = sqrt(144/169) = 12/13.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: The Pythagorean Invariant

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is cos² theta + sin² theta = 1 the single most important identity in engineering? Because it is just the Pythagorean theorem (a² + b² = c²) written on a circ

STEP-BY-STEP APPARATUS SETUP:

1. Pick any angle on the Volvelle (e.g. 30°, 45°, 60°, or an odd angle like 37°).
2. Notice the triangle formed by the origin (0,0), the coordinate point (X, Y), and the X-axis.
3. The base is length cos theta; the height is length sin theta; the hypotenuse is the radius (r = 1).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Unit Circle Volvelle + SC-32 Scientific Hardware Calculator

DATUM TARGET: Pythagorean Theorem: In Right Triangle with Hypotenuse r=1: X² + Y² = r² -> cos² theta + sin² theta = 1

CALCULATOR KEY: [37] [COS] [x²] [37] [SIN] [x²] [+] -> 1.0000

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math, Ch. 9, Lesson 9.3 (Trigonometric Identities)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 5, Sec. 5.1 • CCSS.MATH.HSF.TF.C.8 • TEKS P.4D
GRADE LEVEL BAND:	Grades 10–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 94–102
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 2 (Pythagorean Invariant) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE SUM OF SQUARES AUDIT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Test angle 30°: cos(30°) = sqrt(3)/2; sin(30°) = 1/2.

Step 2: Square both: (sqrt(3)/2)² = 3/4. (1/2)² = 1/4.

Step 3: Add them: 3/4 + 1/4 = 4/4 = 1.0000!

Step 4: Test angle 45°: (sqrt(2)/2)² + (sqrt(2)/2)² = 2/4 + 2/4 = 4/4 = 1.0000!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Pythagorean Theorem: In Right Triangle with Hypotenuse r=1: $X^2 + Y^2 = r^2 \rightarrow \cos^2\theta + \sin^2\theta = 1$

PHYSICAL APPARATUS MANIFEST:

Unit Circle Volvelle + SC-32 Scientific Hardware Calculator

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: TESTING NON-STANDARD ANGLES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Pick an arbitrary angle: 73.4° on your SC-32 calculator.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Calculate cos(73.4°).**
   Expected Physical Finding: **0.2857**
2. **Calculate sin(73.4°).**
   Expected Physical Finding: **0.9583**
3. **Sum their squares: (0.2857)² + (0.9583)².**
   Expected Physical Finding: **0.0816 + 0.9184 = 1.0000 (Exact!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [37] [COS] [x²] [37] [SIN] [x²] [+] -> 1.0000

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
37	X: 37	Arbitrary angle
COS	X: 0.7986	cos(37°)
x²	X: 0.6378	cos²(37°)
37	Y: 0.6378 | X: 37	Angle
SIN	X: 0.6018	sin(37°)
x²	X: 0.3622	sin²(37°)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: Tangent & The Infinite Horizon

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is the tangent line called a tangent line?' (It touches the circle at exactly one point!).
- Ask: 'Can tangent be negative? Where?' (Quadrants II and IV, where slope is downhill).
- Ask: 'At what angle does tan theta equal -1?' (135° and 315°).

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking tangent can't be greater than 1 because sine and cosine can't. Point out that slope can be a
s steep as a cliff!

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"Find an angle whose tangent is equal to its cosine. Can this happen?"
""

OFFICIAL SOLUTION KEY:

sin/cos = cos -> sin = cos² = 1 − sin² -> sin² + sin − 1 = 0 -> sin theta ≈ 0.618 (Golden Ratio angle ≈ 38.17°).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: Tangent & The Infinite Horizon

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does the tangent function shoot up to infinity at 90°? Because tangent is the slope of the angle line, and a vertical line has infinite slope.

STEP-BY-STEP APPARATUS SETUP:

1. Look at the vertical tangent ruler running along the right side of the Volvelle (at the x = 1 tangent line).
2. As you rotate the angle arm, the projection ray extends outward until it intersects this vertical line.
3. The height where it hits the line is the TANGENT of the angle.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Volvelle Slide Ruler with Exterior Tangent Projection Scale (at x = 1 line)

DATUM TARGET: Tangent Invariant: tan theta = Slope of Radial Line = y / x = sin theta / cos theta

CALCULATOR KEY: [89.9] [TAN] -> 572.95 | [90] [TAN] -> ERROR (Division by Zero)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math, Ch. 9, Lesson 9.4 (Tangent Function)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 4, Sec. 4.3 • CCSS.MATH.HSF.TF.B.5 • TEKS P.4C
GRADE LEVEL BAND:	Grades 11–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 102–110
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 6 (Tangent Tangency Line & Asymptote)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE TANGENT ASCENT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Rotate to 0°: Line is horizontal (slope = 0). Reading on tangent ruler: 0.00.

Step 2: Rotate to 45°: Slope is 1 (45° line rises 1 for every 1 run). Reading: exactly 1.000.

Step 3: Rotate to 60°: Slope steepens. Reading: sqrt(3) ≈ 1.732.

Step 4: Rotate toward 90°: Look at the ray: At 85°, it hits 11.43. At 89°, it hits 57.3. At 89.9°, it hits 573! At 90°, the line is parallel to the tangent ruler and NEVER touches it!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Tangent Invariant: tan theta = Slope of Radial Line = y / x = sin theta / cos theta

PHYSICAL APPARATUS MANIFEST:

Volvelle Slide Ruler with Exterior Tangent Projection Scale
(at x = 1 line)

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE DIVISION BY ZERO ASYMPTOTE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Explain why tan(90°) is undefined using sine and cosine.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is sin(90°)?**
   Expected Physical Finding: **1.000**
2. **What is cos(90°)?**
   Expected Physical Finding: **0.000**
3. **Compute tan(90°) = sin(90°) / cos(90°).**
   Expected Physical Finding: **1 / 0 = Undefined (Vertical Asymptote)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [89.9] [TAN] -> 572.95 | [90] [TAN] -> ERROR (Division by Zero)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
89.9	X: 89.9	Near vertical angle
TAN	X: 572.96	Tangent explodes to infinity
90	X: 90	Vertical line
TAN	X: NaN / ERROR	cos(90°) = 0 -> Division by zero asymptote

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 4: The 4-Quadrant Sign Matrix

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does tangent end up positive in Quadrant III if both X and Y are negative?'
- Ask: 'How does finding the reference angle let you solve any trig problem on Earth using only Quadrant I?'
- Ask: 'What happens to cosine and sine on the boundary axes (0°, 90°, 180°, 270°)?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to remember signs by rote memorization. Just ask: 'Is the point to the right or left? Above or below the floor?'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If tan theta > 0 and sin theta < 0, in which quadrant must theta lie?"
""

OFFICIAL SOLUTION KEY:

Quadrant III (where tangent is positive and sine is negative).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 1: The 4-Quadrant Sign Matrix

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does every precalculus student learn the mnemonic 'All Students Take Calculus'? In this lab, rotate through all 4 quadrants to see the sign changes happen in

STEP-BY-STEP APPARATUS SETUP:

1. Rotate the Volvelle through all four quadrants.
2. Quadrant I (Top-Right, 0°–90°): X > 0, Y > 0.
3. Quadrant II (Top-Left, 90°–180°): X < 0, Y > 0.
4. Quadrant III (Bottom-Left, 180°–270°): X < 0, Y < 0.
5. Quadrant IV (Bottom-Right, 270°–360°): X > 0, Y < 0.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Dual-Rotor Volvelle with Quadrant Sign Indicators (QI: +/+, QII: −/+, QIII: −/−, QIV: +/−)

DATUM TARGET: ASTC Rule: QI (All Positive), QII (Sin Positive), QIII (Tan Positive), QIV (Cos Positive)

CALCULATOR KEY: [210] [SIN] -> −0.500 | [210] [COS] -> −0.866

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Additional Math, Ch. 9, Lesson 9.2 (Trig Signs in 4 Quadrants)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Precalculus, Ch. 4, Sec. 4.4 • CCSS.MATH.HSF.TF.A.2 • TEKS P.4B

GRADE LEVEL BAND: Grades 10–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 110–118

HARDWARE UNBOXING BOOKLET: VOLV-01 Booklet: Page 5 (Quadrant Signs) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: THE ASTC VERIFICATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Quadrant I (ALL): cos, sin, and tan are all POSITIVE (+, +, +).

Step 2: Quadrant II (SIN): X is negative, Y is positive. Therefore, ONLY SINE is positive (+). (cos is −, tan is −).

Step 3: Quadrant III (TAN): X is negative, Y is negative. Negative / Negative = Positive! Therefore, ONLY TANGENT is positive (+).

Step 4: Quadrant IV (COS): X is positive, Y is negative. Therefore, ONLY COSINE is positive (+).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
ASTC Rule: QI (All Positive), QII (Sin Positive), QIII (Tan Positive), QIV (Cos Positive)

PHYSICAL APPARATUS MANIFEST:

Dual-Rotor Volvelle with Quadrant Sign Indicators (QI: +/+,
QII: −/+, QIII: −/−, QIV: +/−)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: REFERENCE ANGLES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Every angle in QII, QIII, and QIV has a twin 'Reference Angle' in QI.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is the reference angle for 150° (Quadrant II)?**
   Expected Physical Finding: 180° − 150° = 30°

2. **What is the reference angle for 225° (Quadrant III)?**
   Expected Physical Finding: 225° − 180° = 45°

3. **What is the reference angle for 300° (Quadrant IV)?**
   Expected Physical Finding: 360° − 300° = 60°

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [210] [SIN] –> −0.500 | [210] [COS] –> −0.866

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
210	X: 210	Quadrant III angle
SIN	X: −0.500	Vertical component negative
210	Y: −0.500 | X: 210	Angle
COS	X: −0.866	Horizontal component negative

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Physics Vectors: Force Resolution

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can\'t you just add 40 N + 30 N to get 70 N?'
- Ask: 'How does the unit circle provide the direction vector for any magnitude in physics?'
- Ask: 'How does the SC-32 ->REC and ->POL key automate AC electrical engineering phasor math?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding vector magnitudes directly. Forces in different directions MUST be resolved into components be
fore adding!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"An aircraft flies at 200 mph at a bearing of 45° with a 20 mph tailwind from the South. What is its ground speed?"
""

OFFICIAL SOLUTION KEY:

Aircraft: (141.4, 141.4). Wind: (0, 20). Net: (141.4, 161.4). Ground speed = sqrt(141.4² + 161.4²) ≈ 214.6 mph.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Physics Vectors: Force Resolution

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

When a rocket launches at an angle or a crane pulls on a cable, the force acts diagonally. Engineers must resolve that diagonal pull into horizontal and vertica

STEP-BY-STEP APPARATUS SETUP:

1. Scenario: A cable pulls a crate with a force of 50 Newtons at an angle of 30° above the ground.
2. Set the Volvelle rotor to 30°.
3. Read the unit circle coordinates: (0.866, 0.500).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Unit Circle Volvelle + SC-32 Hardware Polar-to-Rectangular Engine
DATUM TARGET:	Vector Resolution: F_x = |F| · cos theta | F_y = |F| · sin theta
CALCULATOR KEY:	[50] [ENTER] [30] [->REC] -> X: 43.30, Y: 25.00

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math, Ch. 10 (Vectors in 2 Dimensions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 6, Sec. 6.3 • CCSS.MATH.HSN.VM.A.3 • TEKS P.4H
GRADE LEVEL BAND:	Grades 11–12+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4 ('Vectors & Phasors'), pp. 118–130
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 7 (RPN Vector Polar/Rectangular) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: SCALING THE UNIT VECTOR

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: The unit circle vector has length 1: F_unit = (cos 30°, sin 30°) = (0.866, 0.500).

Step 2: Scale the vector by 50 Newtons: Multiply both components by 50.

Step 3: Horizontal pull: F_x = 50 × cos(30°) = 50 × 0.8660 = 43.30 N.

Step 4: Vertical lift: F_y = 50 × sin(30°) = 50 × 0.5000 = 25.00 N.

Step 5: Execute on SC-32: [50] [ENTER] [30] [->REC]. Verify Register X reads 43.30 N and Register Y reads 25.00 N!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Vector Resolution: F_x = |F| · cos theta | F_y = |F| · sin theta

PHYSICAL APPARATUS MANIFEST:

Unit Circle Volvelle + SC-32 Hardware Polar-to-Rectangular E
ngine

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: RESULTANT FORCE SUMMATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Two tugboats pull a barge: Tug A pulls 40 N at 0° (due East). Tug B pulls 30 N at 90° (due North).

FORM A MASTER STATION TASKS (Complete All 3):

1. **Combine vectors: (40, 0) + (0, 30).**
   Expected Physical Finding: **(43, 30) -> Total Vector = (40, 30)**

2. **Calculate resultant magnitude on SC-32: [40] [ENTER] [30] [->POL].**
   Expected Physical Finding: **r = 50.00 N (3-4-5 Triangle!)**

3. **What is the compass heading angle theta?**
   Expected Physical Finding: **theta = 36.87°**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [50] [ENTER] [30] [->REC] -> X: 43.30, Y: 25.00

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
50	X: 50	Vector magnitude |F| = 50 Newtons
ENTER	Y: 50 | X: 50	Stack lift
30	Y: 50 | X: 30	Angle theta = 30 degrees
->REC	Y: 25.00 | X: 43.30	Convert Polar (r, theta) to Rectangular (Fx, Fy)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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